Efficient Multi-Stage Designs for Phase II Clinical Trials Using Swarm Intelligence Optimization Techniques

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Abstract

Multi-stage Phase II clinical trials are increasingly discussed in the literature because they offer strategic advantages by enabling interim analyses that can more accurately inform early termination of the trial if the treatment is ineffective. However, searching optimal multi-stage trials have many challenges. Traditional exhaustive searches are computationally expensive and lack scalability beyond finding trials with two or three stages. Here the Particle Swarm Optimization (PSO) algorithm is modified to generate multi-stage designs for Phase II trials. Specifically, we use PSO to find new multi-stage designs with 3 or more stages for Simon's type designs: optimal design, which minimizes the expected sample size, and minimax design, which minimizes the total sample size, and subject to pre-specified Type I and Type II error constraints. We first show PSO can not only produce the same two-stage and three-stage designs already reported in the literature, it can do so more effectively than traditional exhaustive search methods. We then discuss PSO parameter configurations that can find four-stage and five-stage optimal designs, computing issues and provide a catalog of four or five-stage designs for clinicians.

Keywords: Interim Analysis, Multi-stage Design, Phase IIA Clinical Trial, Particle Swarm Optimization

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1 Introduction

The primary objective of Phase II clinical trials is to assess the efficacy of a novel treatment in a specific patient population, primarily to determine if the drug demonstrates sufficient biological activity to warrant further investigation in Phase III trials. These trials play a crucial role in determining the safety and efficacy of the drug for the treated patients. Our focus is in the design of more flexible Phase II trials, where an overview of such trials is available at Torres-Saavedra and Winter (2022). See for example, Shan et al. (2016); Wu et al. (2021); Jung (2015); Goll and Bauer (2007); Kim and Schell (2019); Cabarrou et al. (2022); Ensign et al. (1994); Pilz et al. (2021); Schultz et al. (1973b); Zang and Lee (2017); Jing et al. (2022)

Multi-stage clinical trials, particularly single-arm trials, are used to conserve resources and designed to terminate a study early if initial results suggest the new drug is ineffective or clearly effective. In the former case, such a trail minimizes the number of patients exposed to a potentially ineffective treatment and in the latter case, prolong the trial unnecessarily. In either case, early termination conserves resources, allowing companies to shift focus to more promising therapeutic candidates, thereby accelerating the development process. There is also an ethical component to early termination as well. Patients should not be treated with a drug as soon as there is evidence to support the finding. This is particular relevant for trials with rare diseases, where patients can move on and participate in other trials. On the other hand, if a drug shows a high response rate early on, there are compelling reasons to continue. Patients benefit from ongoing access to a seemingly effective drug, which could potentially lead to significant health improvements or even a cure. For the company, continuing the trial provides a more comprehensive dataset that is essential for assessing long-term effects and overall efficacy, in addition to supporting regulatory approval and successful market entry. Multi-stage designs enhance manageability by enabling interim decision-making, which conserves resources such as time, cost, effort, and ensures a thorough evaluation of the drug's benefits and risks, and ultimately resulting in a robust clinical and commercial decision-making.

The concept of optimizing multi-stage designs was first proposed by Simon (1989). He focused on two-stage designs for trials with a binary outcome and the goal was to either minimize the expected sample size, or to minimize the maximum sample size. In either case, the sought design controls for the user-specified Type I and Type II errors. Simon's methodologies in the late 1980s laid the groundwork for more structured and statistically rigorous frameworks, particularly for two-stage designs that are now commonly employed in Phase II oncology trials. There are now many variations of Simon's designs that cater to different scenarios, such as trials with missing

data or trials with a delayed response.

Subsequent extensions of Simon's model include Lin and Shih (2004), which allows flexibility by incorporating an adaptive element based on interim results from the first stage. After observing the first stage data, the design can adapt by continuing with either a more optimistic or more conservative target response rate. Chen (1997) introduced a three-stage design by adding an extra decision point to enhance the trial's capability for early termination. This adaptation potentially reduces the necessary number of patients if initial results are not promising. Hanfelt (1999) further expanded the optimal two-stage design to accommodate for dose escalation based on early-stage results, and proposed a four-stage design that maximizes response rates while maintaining control over error rates, thereby enhancing efficiency compared to conducting separate two-stage trials at each dose level. Tan and Xiong (2011) advanced the methodology by optimizing four- and five-stage designs using a sequential testing approach for oncology trials. This method enables continuous data evaluation, reducing patient exposure to ineffective treatments and facilitating the early identification of promising therapies.

Multi-stage trials are currently playing an important role in multi arm multi stage (MAMS) clinical trials. A motivation is that by evaluating multiple experimental arms against a common control within one trial, the number of patients required in the trial is reduced compared to a series of trials testing each experimental arm separately against control. It is also thought that treatments which are ineffective can be eliminated from a MAMS trial earlier more efficiently instead of having to conduct several two-arm trials to determine its effectiveness; see details in Wason et al. (2016). A multi-arm multi-stage trial can supposed find effective treatments faster but it requires considerable financial resources; so a careful design of a MAMS trial is critical. Our work can enhance the design of such trials by optimally designing the multi-stage aspects of a MAMS trial.

Simon's two-stage design continues to play a pivotal role in Phase II clinical trials, especially for evaluating new treatments' efficacy. All statistical packages have options to find Simon's two-stage designs. The methodology is robustly supported by various online tools and platforms, such as the MD Anderson Cancer Center's ShinyApp (Zhang and Yuan, 2020), SWOG Cancer Research Network's two-stage program (SWOG, 2024), Clinical Biostats' calculator (Clinical Biostats, 2023), and resources from UNC Lineberger (Ivanova, 2012) and Alira Health (Alira Health, 2024). Additionally, NCSS incorporates a Simon's design calculator into their commercial software (NCSS Software, 2024). Open-source code is also available for R users through packages (Sechan and Whiting, 2023; Zhan, 2024) from CRAN to generate Simon's design. These resources underscore the widespread accessibility and usage of Simon's two-stage design, particularly for

preliminary efficacy assessments within the research community.

However, there remains a pronounced lack of resources for creating trial designs that extend beyond two stages. Developing multi-stage designs typically involves specialized exhaustive search methods that search through every possible combination of cohort sizes and cut-off levels at each stage to meet pre-specified error constraints. As clinical trials grow in complexity and the demand for more adaptable designs escalates, the need for tools that support extended multi-stage clinical trial designs becomes increasingly critical. Current methods for finding a trial with 3 or more stages are scarce and generally do not perform well. A key issue is that it is very computational demanding to search for optimal multi-stage designs because of the numbers of both discrete and continuous variables required to be optimized. Hence more efficient algorithms to streamline the optimization of multi-stage designs in Phase II clinical trials are needed to advance clinical research. The development of such innovative tools would markedly enhance trial design flexibility and efficiency, and align better with the diverse and evolving needs of research outcomes.

1.1 Nature-inspired metaheuristic algorithms

We propose nature-inspired metaheuristic algorithms to tackle the computation complexity in finding multi-stage clinical trial designs. This class of algorithms is commonly used in engineering, computer science and artificial intelligence to tackle hard optimization problems but rarely seen in the clinical trial research. Their meteoric rise in popularity in use in industry and academia is well documented in Whitacre (2011a,b). Some key reasons are that they require no technical assumptions for them to work well, their relatively simple codes are widely available, and they are easy to implement across various platforms. An intriguing aspect is that these algorithms invariably do not have rigorous proofs of convergence to the global optimum and yet they tend to work well for tackling all types of optimization problems in practice.

There are many such algorithms and they are broadly divided into two classes: swarm based and evolutionary based. They are all somewhat competitive in their own ways, and motivated differently by nature or animal behavior, as the name suggests. Some common features among them are that they have random elements in the algorithms to make the algorithm stochastic and help them get out of a local optimum. All operate using the exploration and exploitation principle, and each has its own modifications, along with its pros and cons. They also have a few tuning parameters that require the user to provide inputs; all algorithms come with default values for the tuning parameters but some are more sensitive to their values. There are many

monographs about such algorithms and they include Poli et al. (2007); Yang (2010); Siarry (2016). Some good papers with an overview and applications of metaheuristics are Gogna and Tayal (2013); Oliveira (2015); Desale et al. (2015); Korani and Mouhoub (2021).

In what is to follow, we focus on an exemplary swarm based nature-inspired metaheuristic algorithm called particle swarm optimization (PSO), and apply it to find Phase II designs for clinical trials with 3, 4 and 5 stages. Qiu et al. (2014) is likely the first to use PSO to find locally optimal designs for estimating parameters for nonlinear models in some biomedical problems. The models were relatively simple and so they were able to confirm PSO results with those that were derived analytically. When models are complex and optimal designs cannot be found analytically, algorithms are useful and it has been shown that PSO was able to find optimal designs that traditional algorithms cannot.

The celebrated Simon's two-stage designs for a Phase II trial are still popular, but there are also criticisms of the designs. Many modifications of the original version have emerged to enhance the original designs. For example, one criticism is that the original version unrealistically assumed the response rate is known and posited in the alternative hypothesis. Lin and Shih (2004) introduced two alternative hypotheses, instead of just one in Simon's origin design setup, to allow two postulated values of the efficacy rate of the drug. Following Simon's method, they employed a greedy search and could not obtain all Phase II designs because of the complexity of the optimization problem and the long computing time involved. In what may be the first use of metaheuristics to tackle a clinical trial design problem, Kim and Wong (2018) used a discrete PSO (DPSO) algorithm to search for a two-stage adaptive designs for the same problem with three alternative hypotheses to further capture the uncertainty of the hypothesized efficacy of the drug, and only one of them will be tested, depending on the results in the first stage. The optimization problem is complex because of the multiple type I and II constraints imposed on the problem and the multiple integer-valued variables to be optimized are also necessarily ordered.

Since then, there have been many successes of using PSO to find optimal designs for increasingly complex problems. For example, locally optimal designs can be sensitive to nominal values and efficiencies can drop by a lot if they are mis-specified. A minimax strategy can be helpful where one designs assuming the worst case scenario. If a set of plausible values for the model parameters is available, this approach would first find the nominal values that would give the design criterion the worst value. Then a design is determined that would make the worst value as small as possible. For instance, Chen et al. (2017) used the NestedPSO algorithm, a modified version of PSO, in Chen et al. (2015) and found a standardized minimax D-optimal design estimating

parameters in a few enzyme kinetic inhibition models. Further, motivated by a study by Slob (2002), who claimed that many univariate continuous outcomes in toxicology can be modelled by one of five nonlinear models, Chen et al. (2020) modified PSO to find different optimal discrimination designs to identify the best fitting models among the five models, depending whether errors are assumed to be normally distributed or not. We are further encouraged by recent continuing successes of PSO in finding various types of challenging optimal designs, and they include Xu et al. (2019); Liu et al. (2021); Stokes et al. (2024). Xu et al. (2019) used an evolutionary based algorithm called differential evolution and found locally optimal designs for estimating parameters in a logistic model with multiple interacting factors, Liu et al. (2021) found optimal designs for estimating response surface with a global protection for hierarchical linear models under a non-differentiable criterion, and Stokes et al. (2024) found optimal designs for order-of-addition experiments. The latter is used in drug combination studies that additionally incorporate the order of addition of the drugs in the design criterion. In what is to follow, we first review the setup for designing a multi-stage study in the next section before we use metaheuristics and create more flexible and new optimal exact designs for clinical trials with multi-stages. .

Our aim is to develop a useful particle swarm optimization algorithm for finding more flexible multi-stage designs for Phase II clinical trials subject to user-specified error constraints. We also investigate the effectiveness of PSO under its various settings of tuning parameters. The rest of the article is organized as follows. Section 2 introduces the theoretical development of the Phase II multi-stage trial designs. We also construct the recursive algorithm for calculating the probability of early termination at a given number of stages. Section 3 describes PSO and provides a complete tutorial on how to adopt PSO in finding multi-stage designs. The tutorial covers the PSO particles' formation, design parameters' reparameterization, and constraints' handling. We present the more flexible four and five-stage designs found by PSO in Section 4 after we showed our modified PSO algorithm is able to produce the same two-stage designs in published work, and a conclusion in the section 5. Section 5.

2 Optimal Multi-stage Designs

The Phase II multi-stage clinical trial designs focus on testing the null hypothesis $H_0 : p \leq p_0$, which posits that the true response rate p is less than or equal to a specified ineffective level p_0. Correspondingly, we consider an alternative hypothesis $H_1 : p \geq p_1$, where the true response rate is at least a desired effective level p_1. When the null hypothesis H_0 is true, the design

requires that the probability of incorrectly concluding that the drug is effective be less than α, controlling the Type I error rate. Conversely, under the alternative hypothesis H_1, we require that the probability of incorrectly rejecting the drug due to perceived ineffectiveness be less than β. This ensures that the power of the test, the probability of correctly rejecting H_0 when H_1 is true, is at least $1 - \beta$.

For a K-stage design, where $K \geq 2$, the number of patients enrolled at each stage is denoted by $n_1, n_2, \ldots, n_K$, respectively. Consequently, the total sample size is given by $N_{max} = \sum_{k=1}^{K} n_k$. The drug is rejected, and the study is terminated if:

- r_1 or fewer responses are observed at the first stage, or
- $r_1 + r_2$ or fewer responses are observed at the second stage, or
- $r_1 + r_2 + r_3$ or fewer responses are observed at the third stage, or
- $\ldots$
- $\sum_k r_k$ or fewer responses are observed at the Kth stage.

Following Simon (1989) and Chen (1997), the probability of early termination (PET) after the first stage is:

$$PET_1(p) = B(r_1, n_1, p) = \sum_{x \leq r_1} b(x, n_1, p),$$

where $b(x, n, p)$ is the probability mass function and $B(x, n, p)$ is the cumulative distribution function of a Binomial random variable with parameters n and p, where n is a positive integer and $p \in [0, 1]$. For $k = 2, \ldots, K$, Liu (2022) showed that the PET at the kth stage has the following general analytical form:

$$PET_k(p) = \underbrace{\sum_{x_1=L_1}^{U_1} \sum_{x_2=L_2}^{U_2} \cdots \sum_{x_j=L_j}^{U_j} \cdots \sum_{x_{k-1}=L_{k-1}}^{U_{k-1}}}_{k-1 \text{ Summations}} \left[\prod_{s=1}^{k-1} b\left(x_s, n_s, p\right)\right] B\left(r_k - \sum_{s=1}^{k-1} x_s, n_k, p\right), \tag{1}$$

where

$$U_j = \min\left[n_j, r_k - \sum_{s=1}^{j-1} x_s\right] \quad \text{and} \quad L_j = r_j + 1 - \sum_{s=1}^{j-1} x_s. \tag{2}$$

For the final stage, the Kth stage, the value $PET_K(p)$ represents the probability of rejecting the treatment at the end of the study. Throughout the trial, the total probability of rejecting the drug in the study is:

$$P(\text{Reject the drug} \mid p) = \sum_{k=1}^{K} PET_k(p). \tag{3}$$

Thus, the Type I and Type II errors are $1 - P(\text{Reject the drug} \mid p_0)$ and $P(\text{Reject the drug} \mid p_1)$, respectively.

The Phase II trial commonly aims to reduce the number of patients receiving an ineffective treatment. Therefore, due to ethical considerations, it is important to minimize the expected sample size under the null hypothesis, $E(N \mid H_0)$, which is expressed as

$$E(N \mid H_0) = n_1 + \sum_{k=2}^{K} n_k \left[1 - \sum_{s=1}^{k-1} PET_s(p) \right]. \tag{4}$$

The design that minimizes (4) is called the *optimal design*. Another widely used design seeks to minimize the maximum sample size while satisfying error constraints. If there are multiple designs are feasible and have minimized maximum sample size, the secondary criterion is used to break ties. Among those feasible designs, the *minimax design* has the smallest value of (4).

Equation (4) involves the computation of PET_k for $k = 1, \ldots, K-1$. For each k, this requires recursive summation of the binomial probabilities. Following Schultz et al. (1973a), Algorithms 1 and 2 illustrate the recursive algorithm for computing PET_k.

Algorithm 1 kStagePET(n, r, p, k): The main function of PET at stage k, $1 \leq k \leq K$

```
Input the following: k, the number of stages; n, the vector of sample size at each interim
  look and the end; r, the vector of decision boundary; p, response probability.
if k == 1 then
    Return B(r, n, p)
else
    Initialize j = 1, x = 0_k
    Return exactProb(j, x, n, r, p, k), see Algorithm 2.
end if
```

Algorithm 2 $exactProb(j, \mathbf{x}, n, r, p, k)$

```
if j < k then
    Compute L_j, U_j by (2)
    for v from L_j to U_j do
        Set x_j ← v
        PET ← PET + exactProb(j + 1, x, n, r, p, K)
    end for
else
    Return [∏_{s=1}^{k-1} b(x_s, n_s, p)] B(r_k − Σ_{s=1}^{k-1} x_s, n_k, p)
end if
```

Exhaustive search has traditionally been employed to identify optimal or minimax multi-stage designs (Simon, 1989; Chen, 1997; Hanfelt, 1999). This method involves enumerating all possible combinations $(n_1, \ldots, n_K)$ that satisfy $\sum n_k = N_{max}$. For each combination, the cut-off points at each stage are determined, with r_k ranging from 0 to n_k for $k = 1, \ldots, K$. The

algorithm iterates through all combinations of cohort sizes and cut-off points, selecting feasible designs that meet predefined Type I and II error constraints, α and β. Among these feasible designs, the one that best meets the specified criterion is selected as optimal.

While the exhaustive search method remains feasible for two- and three-stage designs given current computing capabilities, as described in Simon (1989) and Chen (1997), it becomes impractical for designs with $K > 3$ due to the exponential increase in the number of candidate designs (billions to trillions) and the associated computational costs.

To reduce the computational burden, certain strategies can be implemented to narrow the set of candidate designs. For instance, Hanfelt (1999) applied an additional constraint related to the power of detecting an effective low dose in the first stage of a four-stage trial. This constraint helps to eliminate infeasible pairs of (n_1, r_1), thereby reducing the overall number of combinations (n_k, r_k) for $k = 1, 2, 3, 4$ that need to be considered in their exhaustive search. Nonetheless, these modifications are specific to particular design problems, and as of this writing, no systematically efficient method exists for generating multi-stage designs.

3 Swarm-based Algorithm for Finding the Optimal Multi-stage Designs

Simon found the two-stage optimal designs using a greedy approach by considering all possible designs and evaluating them one by one. When the trial has 3 or more stages, $K \geq 2$, the number of candidate designs is too large for such a brute-force search to work well. In practice, the search for a multi-stage design for a clinical trial may begin with a minimum of m_1 patients in the first stage, and at the kth, recruit m_k additional patients in each subsequent stage. Assuming $m_1 = 5$ and $m_k = 1$ for $k > 1$, Table 1 presents the total number of candidate designs considered during the exhaustive search for various maximal sample sizes $N_{max} = 30, 40, 50, 60$ and 70.

When the maximal sample size N_{max} is not predetermined and the objective is to identify the optimal design among all possible values of N_{max} from 30 to 70, the computational load of the exhaustive search increases significantly. Under this scenario, the number of candidate designs amounts to 1,077,892 for $K = 2$, 200,388,651 for $K = 3$, 20,179,606,693 for $K = 4$, and 1,299,302,807,995 for $K = 5$. Clearly, the greedy search used in Simon's original paper is not applicable and even if it did, the computation time will be excessive. To this end, we resort to nature-inspired metaheuristic algorithms, which are variously known as last resort optimization methods or general purpose optimization methods, to conduct the search for an optimal multi-

Table 1: The total number of candidate K-Stage designs at given maximal sample size N_{max}. The minimal sample size at the first stage is $m_1 = 5$, and, that at the kth stage is $m_k = 1$ for $k > 1$.

K	$N_{max} = 30$	$N_{max} = 40$	$N_{max} = 50$	$N_{max} = 60$	$N_{max} = 70$
2	5,000	11,725	22,650	38,775	61,100
3	251,230	1,045,772	3,132,789	7,659,531	16,303,248
4	6,278,540	46,287,736	214,365,492	746,328,033	2,140,111,584
5	95,230,465	1,242,500,556	8,873,200,457	43,873,633,518	169,076,267,464

stage trial. We focus on one exemplary member called particle swarm optimization (PSO) with a well documented record of success for finding an optimum or nearly optimal solution across disciplines.

3.1 Particle Swarm Optimization

Particle Swarm Optimization (PSO) is a metaheuristic optimization method introduced by Eberhart and Kennedy (1995). Inspired by the collective behavior of animals, such as flocks of birds searching for food, this nature-inspired algorithm simulates the coordinated movements of these birds as they search for an optimal solution, akin to finding food on the ground. Throughout the process, particles—representing candidate solutions—communicate and adjust their velocities and positions iteratively until they converge on an optimum or the algorithm is terminated by a user-specified stopping criterion.

The PSO algorithm begins by randomly generating a swarm of N particles in the given design space. Each particle's position is represented by $\xi_i^{(t)}$ at iteration t, and the particles move through the solution space, guided by two key concepts: the local best and the global best. The local best, denoted as $\xi_{i*}^{(t-1)}$, is the best solution found by the ith particle up to iteration $t-1$. The global best, denoted as $\xi_g^{(t-1)}$, is the best solution found by the entire swarm up to iteration $t-1$. The movement of each particle is governed by its velocity $V_i^{(t)}$ at iteration t, which is updated using the following equation:

$$V_i^{(t)} = \omega^{(t)} V_i^{(t-1)} + c_1 R_1 \otimes \left[\xi_{i*}^{(t-1)} - \xi_i^{(t-1)}\right] + c_2 R_2 \otimes \left[\xi_g^{(t-1)} - \xi_i^{(t-1)}\right], \tag{5}$$

where $\omega^{(t)}$ is the inertia weight controlling the influence of the previous velocity, c_1 and c_2 are the cognitive and social learning factors, respectively, and R_1 and R_2 are random vectors with components uniformly drawn from $[0, 1]$. The Hadamard product, denoted by $\otimes$, is used for the element-wise multiplication of vectors. After updating the velocity, the new position of the ith

particle at iteration t is calculated as:

$$\xi_i^{(t)} = \xi_i^{(t-1)} + V_i^{(t)}. \tag{6}$$

The choice of parameters is crucial for the performance of the PSO algorithm. The inertia weight $\omega^{(t)}$, which typically decreases linearly from 0.95 to 0.2 over the first 80% of the iterations, and the learning factors c_1 and c_2, which have default values equal to 2 in Eberhart and Shi (2000). These parameters ensure that the particles explore the solution space efficiently, balancing exploration and exploitation. The swarm size N, the number of particles, is typically determined by the complexity of the optimization problem. A common stopping criterion is either a pre-specified number of iterations, a time limit, or a maximum number of function evaluations. Algorithm 3 provides the pseudo code for the basic PSO algorithm. For more details on PSO, please refer to Yang (2010).

Algorithm 3 Pseudo code of the PSO for Minimization

1: Define the objective function, denoted by $f(\xi)$, and **Input** the following: the swarm size N, along with values for the tuning parameters.

(1.1) Generate *initial particles* $\xi_i^{(0)}$ *and velocities* $V_i^{(0)}$, $i = 1, \ldots, N$.

(1.2) Calculate *objective function values* $f(\xi_i^{(0)})$ *for each* i.

(1.3) Initialize *the local and global best designs,* $\xi_{i^*}^{(0)} = \xi_i^{(0)}$ *and* $\xi_g^{(0)} = \min_i \ \xi_{i^*}^{(0)}$.

2: At the tth iteration, **do**

(2.1) Calculate *particles' velocities* $V_i^{(t)}$ *by (5).*

(2.2) Update *particles* $\xi_i^{(t)}$ *by (6).*

(2.3) Calculate *objective function values* $f(\xi_i^{(0)})$ *for each* i.

(2.4) Update *the local best designs* $\xi_{i^*}^{(t)} = \min_{s=0,1,\ldots,t} \xi_i^{(s)}$.

(2.5) Update *the global best design* $\xi_g^{(t)} = \min_i \ \xi_{i^*}^{(t)}$.

3: **Output** the final global best design ξ_g and $f(\xi_g)$.

Given the number of stages, K, the optimal K-stage designs have the structure

$$\psi = \{r_1, n_1, r_2, n_2, \ldots, r_K, n_K\}$$

where the r_k, n_k are integers and $0 \leq r_k \leq n_k$ for $k = 1, 2, \ldots, K$, and, $\sum_k n_k = N_{max}$ the total number of patients. To find the best K-stage design using PSO, it is helpful to reparameterize

the design ψ as the PSO's particle. Our approach of the reparameterization aims to identify the n_k's by searching their proper proportion to the N_{max} and to identify each r_k by searching its proper proportion to the corresponding n_k. Let the PSO's particle be a $2K$-dimensional vector on the continuous domain

$$\xi = (\Omega, \theta_1, \ldots, \theta_{K-1}, \phi_1, \ldots, \phi_K). \tag{7}$$

The first component $\Omega \in [\Omega_L, \Omega_U]$ is the reparameterized total number of patients and $N_{max} = \lfloor \Omega + 0.5 \rfloor$. Components $\theta_k \in [0, \pi/2]$, $k = 1, \ldots, K-1$, are the angles of the polar coordinate system in the K-dimensional space, that are used as the reparameterized sample size at each stage to address the sum constraint on the n_k's. For a given set of θ_k's, by the following transformation

$$\begin{aligned}
\sqrt{\omega_1} &= \cos(\theta_1) \\
\sqrt{\omega_i} &= \prod_{k=1}^{i-1} \sin(\theta_k) \cos(\theta_i), \ i = 2, \ldots, K-1 \\
\sqrt{\omega_K} &= \prod_{k=1}^{K-1} \sin(\theta_k)
\end{aligned}$$

we have $\sum \omega_k = 1$, and then we compute $n_k = \lfloor \omega_k \times N_{max} + 0.5 \rfloor$. If there are constraints on the minimal number of patients at each stage, denoted by $m_1, m_2, \ldots, m_K$, then $n_k = \lfloor \omega_k \times (N_{max} - \sum_k m_k) \rfloor + m_k$. Finally, components $\phi_k \in [0, 1]$, $k = 1, \ldots, K$, are the reparameterized cutoff of rejecting a drug and we transfer them by $r_k = \lfloor \phi_k \times n_k + 0.5 \rfloor$. If there are constraints on the cutoff increment from stage 1 to stage K, denoted by $q_1, q_2, \ldots, q_K$, then $r_k = \lfloor \phi_k \times (n_k - q_k) + 0.5 \rfloor + q_k$. The default setup for the q_ks is a zero-valued vector. If we are aiming for a meaningful multi-stage design that does not degenerate to a design with fewer stages, the setup for the q_k's would be $(q_1, q_2, \ldots, q_K) = (0, 1, 1, ..., 1)$ indicating that the cutoff boundary must increase by one patient for every stage.

Following ideas in Simon (1989), the optimal K-stage designs are developed to minimize the expected sample size when the treatment is ineffective. The primary objective of this optimization is to reduce the number of patients exposed to an ineffective treatment, all while maintaining control over Type I and Type II error rates. In PSO, the goal of finding an optimal K-stage design involves minimizing the objective function

$$f_{opt}(\xi) = E(N|H_0) + M \times \max\{I(\text{t1e} > \alpha), I(\text{t2e} > \beta)\}, \tag{8}$$

where $E(N|H_0)$ represents the expected sample size under the null hypothesis, and M is a

penalty term applied when the design does not satisfy the specified Type I (α) and Type II (β) error constraints. The indicator function $I(\cdot)$ enforces these error constraints.

Additionally, Simon (1989) aimed to minimize the maximum sample size while still adhering to these error constraints. Therefore, the task of identifying the minimax K-stage design requires minimizing the objective function

$$f_{mm}(\xi) = \left[N_{max} + \frac{1}{N_{max}} E(N|H_0)\right] + M \times \max\{I(\text{t1e} > \alpha), I(\text{t2e} > \beta)\}. \tag{9}$$

Here, the primary focus of minimizing f_{mm} is on reducing N_{max}, the maximum sample size. Once N_{max} is minimized, the goal is to identify the most ethical design, characterized by the smallest $E(N|H_0)$ at the fixed N_{max}.

In both objective functions (8) and (9), M is a very large user-selected positive constant that serves as a significant penalty for designs that do not meet the Type I error and power requirements. A typical value is $M = 10^8$ to strongly enforce the error constraints.

To evaluate the optimality of the given design ψ, we define the $E(N|H_0)$-efficiency by

$$\text{Eff}_{opt}(\psi) = \frac{f_{opt}(\psi^*)}{f_{opt}(\psi)} \times 100\% = \frac{E_{\psi^*}(N|H_0)}{E_{\psi}(N|H_0)} \times 100\% \tag{10}$$

where ψ^* is the best design found in the literature and the current numerical experiments. This efficiency value represents the relative cost of a given design against to the optimal design. For example, a design with 50% efficiency indicates that the expected number of patients of this design is doubled comparing to that of the optimal design.

4 Numerical Results

We first show in the first subsection that our modified algorithm produces the same results in the published literature for 2 and three-stage designs. Since choice of values for the tuning parameters can have an impact of the performance of a metaheuristic algorithm, we investigate their effects on the search capability of our modified algorithm. Subsection lists various optimal multi-stage designs for Phase 2 trial under various assumptions and configurations of the design problems. The tables report the two types of optimal designs sought in Simon's original paper, and present the PSO-generated multi-stage designs in a similar way as in the original version, with obvious extensions.

4.1 Revisiting two- and three-stage designs

We first revisit the results of two-stage designs as detailed in Simon (1989). We search for two-stage designs with total sample sizes ranging between 10 and 120 patients, with minimum sample sizes at each stage set at $(m_1, m_2) = (2, 1)$. The trial requirements are specified as $(p_0, p_1, \alpha, \beta) = (0.2, 0.4, 0.1, 0.1)$. We employ Particle Swarm Optimization (PSO) with 128 particles for 400 iterations. Tables 2 and 3 display the resulting PSO-generated two-stage optimal and minimax designs, respectively. PSO consistently identifies the optimal and minimax designs, as initially shown in Simon (1989), in 100 replications. In instances where the algorithm does not achieve the global minimax design, as recorded in Table 3, the resulting designs maintain the same maximum sample size, $N_{max} = 36$, but exhibit a higher $E(N|H_0)$ than that found by Simon (1989), with the lowest $E(N|H_0)$-efficiency at 96.74%.

Subsequently, we demonstrate that PSO can reproduce three-stage optimal and minimax designs as described in Chen (1997). Following the previous parameters, we explore three-stage designs with total sample sizes between 10 and 120 patients, setting the minimum sample sizes at each stage to $(m_1, m_2, m_3) = (2, 1, 1)$. Given the increased complexity, we employ PSO with 256 particles over 800 iterations. In this setup, PSO successfully locates the optimal and minimax designs at least 67% of the time across 100 independent runs. For runs not achieving the optimal design, as seen in Table 4, the resulting designs achieve a commendable $E(N|H_0)$-efficiency of 99.40%. For instances falling short of the global minimax design, as shown in Table 5, the resulting designs maintain an $N_{max} = 36$, with $E(N|H_0)$-efficiencies ranging from 96.18% to 98.83%.

Finally, we present results where PSO identifies new three-stage optimal and minimax designs that outperform those documented in Chen (1997). With trial requirements set at $(p_0, p_1, \alpha, \beta) = (0.4, 0.6, 0.05, 0.2)$, we run PSO with 256 particles for 800 iterations to identify corresponding optimal and minimax designs. Table 6 reveals that PSO discovers a new optimal three-stage design achieving 100.16% $E(N|H_0)$-efficiency relative to the results shown in Chen (1997), with this new optimal design identified in 89 out of 100 runs. In Table 7, the newly identified minimax designs require a total of 39 patients, compared to the 41 patients required by the designs in Chen (1997), making the PSO-generated designs more cost-effective.

Table 2: PSO-generated two-stage optimal designs using 128 particles and 400 iterations $(p_0, p_1, \alpha, \beta) = (0.2, 0.4, 0.1, 0.1)$

Method	Resulting Design	T1E	Power	E(N|H0)	#(times)
Simon (1989)	3/17, 10/37			26.022	
PSO	3/17, 10/37	0.095	0.903	26.022	100/100

Table 3: PSO-generated two-stage minimax designs using 128 particles and 400 iterations $(p_0, p_1, \alpha, \beta) = (0.2, 0.4, 0.1, 0.1)$

Method	Resulting Design	T1E	Power	E(N|H0)	#(times)
Simon (1989)	3/19, 10/36			28.263	
PSO Top 1	3/19, 10/36	0.086	0.902	28.263	98/100
PSO Top 2	4/22, 10/36	0.086	0.903	28.400	1/100
PSO Top 3	5/25, 10/36	0.086	0.904	29.216	1/100

Table 4: PSO-generated three-stage optimal designs using 256 particles and 800 iterations $(p_0, p_1, \alpha, \beta) = (0.2, 0.4, 0.1, 0.1)$

Method	Resulting Design	T1E	Power	E(N|H0)	#(times)
Chen (1997)	1/10, 6/26, 11/43			23.860	
PSO Top 1	1/10, 6/26, 11/43	0.100	0.901	23.860	67/100
PSO Top 2	2/14, 7/29, 10/38	0.098	0.901	24.003	33/100

Table 5: PSO-generated three-stage minimax designs using 256 particles and 800 iterations $(p_0, p_1, \alpha, \beta) = (0.2, 0.4, 0.1, 0.1)$

Method	Resulting Design	T1E	Power	E(N|H0)	#(times)
Chen (1997)	2/16, 5/26, 10/36			26.413	
PSO Top 1	2/16, 5/26, 10/36	0.086	0.901	26.413	95/100
PSO Top 2	3/19, 6/29, 10/36	0.086	0.901	26.726	1/100
PSO Top 3	3/20, 4/22, 10/36	0.086	0.902	27.462	4/100

Table 6: PSO-generated three-stage optimal designs using 256 particles and 800 iterations $(p_0, p_1, \alpha, \beta) = (0.4, 0.6, 0.05, 0.2)$

Method	Resulting Design	T1E	Power	E(N|H0)	#(times)
Chen (1997)	3/ 9, 10/23, 23/46			21.740	
PSO Top 1	3/ 9, 11/24, 25/51	0.049	0.803	21.706	89/100
PSO Top 2	3/ 9, 10/23, 23/46	0.049	0.801	21.739	8/100
PSO Top 3	6/14, 15/33, 25/51	0.049	0.801	22.411	1/100
PSO Top 4	7/16, 16/37, 23/46	0.048	0.800	23.502	2/100

Table 7: PSO-generated three-stage minimax designs using 256 particles and 800 iterations $(p_0, p_1, \alpha, \beta) = (0.4, 0.6, 0.05, 0.2)$

Method	Resulting Design	T1E	Power	E(N|H0)	#(times)
Chen (1997)	4/12, 11/25, 21/41			23.170	
PSO Top 1	7/19, 12/26, 20/39	0.050	0.800	25.130	18/100
PSO Top 2	7/18, 16/33, 20/39	0.050	0.801	25.226	24/100
PSO Top 3	9/22, 12/26, 20/39	0.050	0.800	26.037	7/100
PSO Top 4	5/16, 15/31, 20/39	0.050	0.803	27.081	1/100
PSO Top 5	10/24, 15/31, 20/39	0.050	0.802	27.449	8/100

4.2 Tuning PSO Parameters for Finding Multi-stage Design

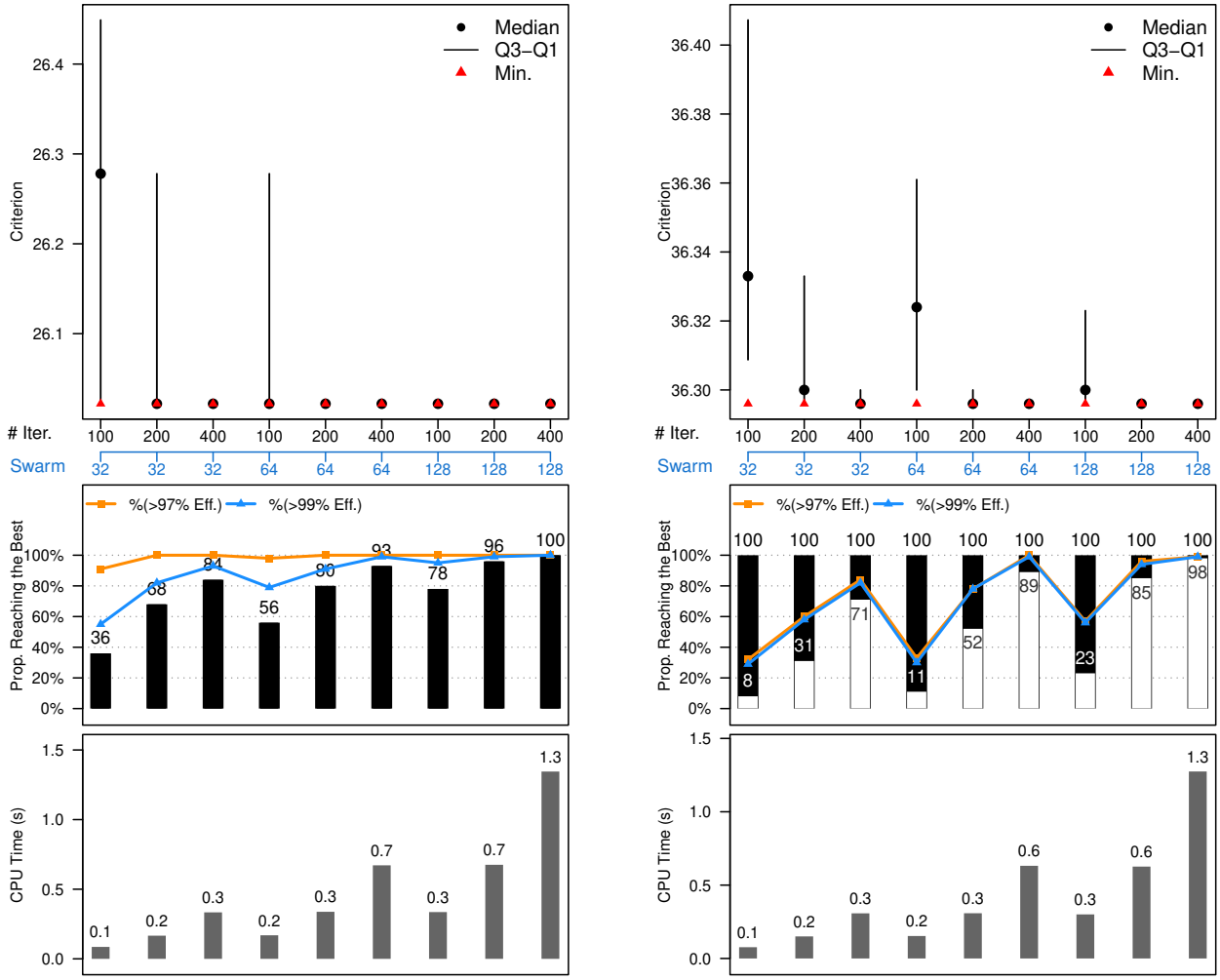
Nature-inspired metaheuristic algorithms have tuning parameters and the user can provide values for them or use their default values. Some algorithms are very sensitive to the values, meaning that its performance can critically depend on it having the right values for the tuning parametersm including the swarm size and number of iterations allowed for the search. In this subsection, we investigate efficient PSO configurations in the algorithm for finding the K-stage optimal and minimax designs. To fix ideas, we consider a selected case when we have $(p_0, p_1) = (0.2, 0.4)$ and $(\alpha, \beta) = (0.1, 0.1)$. The swarm size is set to 2^{k+5}, where $k = 1, \ldots, K$ and the maximal number of iterations is set to 100×2^k, where $k = 0, \ldots, K$. Other tuning parameters in the PSO configuration are set to their default values. For each PSO configuration, the experiments are conducted with 100 replications in a computer with an Intel Core(TM) i7-13700 2.10 GHz CPU and equipped with 128 GB RAM. Below, we summarize the resulting criterion values, the frequencies of finding the best design over replications, and the average computing time.

Figure 1 displays the results of the numerical experiments for the case of finding two-stage optimal and minimax designs. The figure shows the PSO's performances based on different swarm sizes, 32, 64 and 128 particles, and, different number of iterations, 100, 200 and 400 iterations. First, subfigure 1 (a) summarizes the results of finding the two-stage optimal design. Its upper panel summarizes the 100 resulting designs' criterion values found by PSO. The dots represent the median, the straight line shows the range from the first and third quantiles, and the triangles indicate the minimal value. Its middle panel uses bars to present the frequency of finding the optimal design, with minimal $E(N \mid H_0)$, in the 100 replications. We also present the frequencies of the PSO-generated designs meeting some certain level of $E(N|H_0)$-efficiency which is defined in (10). The orange line and the blue line indicate the frequency that the resulting design achieves at least 97% efficiency and 99% efficiency to the overall best design, respectively. The bottom panel presents the average computing time in seconds of the PSO search under each PSO configuration of swarm size and number of iterations. Secondly, subfigure 1 (b) summarizes the results of finding the two-stage minimax design. The interpretations for its upper and bottom panels follow subfigure 1 (a). For interpreting its middle panel, we show two bars to present the performance on finding the minimax design. The dark bars present the frequencies of finding the design with minimal total sample size, N_{max}, in the 100 replications. The light bars present the frequency of finding minimax design with minimal $E(N \mid H_0)$ among all successes cases of identifying the minimal N_{max}. The orange and blue lines in this middle

panel are the frequencies of the PSO-generated minimal-N_{max} designs reaching 97% and 99% $E(N|H_0)$-efficiencies comparing to the overall best minimax design, respectively.

We present the PSO's performances using different swarm sizes and number of iterations for finding three-stage designs in Figures 2 and 3, for finding four-stage designs in Figures 4 and 5, and, for finding five-stage designs in Figures 6 and 7. For four- and five-stage designs, we mean the optimal and minimax designs to be the overall best ones found in our massive numerical experiments.

With more particles in the swarm and more iterations, PSO costs more computing time with higher chance to generate the design closed to the over best one. For three-stage designs, we recommend running PSO with 256 particles and 800 iterations for finding the optimal and minimax designs with low computational cost in 18 seconds. For the designs with more stages, the computational cost increases exponentially. Therefore, there is a trade-off between design's quality and computing time. For example, PSO has 78% chance to identify the four-stage optimal design if running with 256 particles and 800 iterations. To improve the chance of finding the best design, the user can double the settings on either swarm size or number of iterations. As shown in Figure 4, the chance is 91% if increasing the number of iterations to 1600 iterations, and, the chance is 82% if increasing the swarm size to 512 particles. For both options, the computing time is doubled. On the other hand, suppose a design with relatively high $E(N|H_0)$-efficiency is good enough for the study, PSO can produce the highly efficient designs quickly. In Figure 4, we find that for 256 particles and 800 iterations, PSO can identify the designs with at least 97% and 99% $E(N|H_0)$-efficiencies almost every implementation in our experiment.



(a) Optimal Design

(b) Minimax Design

Figure 1: The performance of the PSO algorithm for finding a two-stage design using different configurations of swarm size and number of iterations. For each configuration, we conduct 100 times of replications starting with different initial swarms. (a) The performance of PSO in finding two-stage optimal design. The upper panel summarizes the 100 resulting designs' criterion values found by PSO. The dots represent the median, the straight line shows the range from the first and third quantiles, and the triangles indicate the minimal value. The middle panel uses bars to present the frequency of finding the optimal design, with minimal $E(N \mid H_0)$, in the 100 replications. In addition, the orange line and the blue line indicate the frequency that the resulting design achieves at least 97% efficiency and 99% efficiency to the overall best design, respectively. The bottom panel presents the average computing time in seconds of the PSO search under each configuration; (b) The performance of PSO in finding two-stage minimax design. The interpretations for the upper and bottom panels, and the orange and blue lines in the middle panel, follow subfigure (a). In the middle panel, there are two bars. The dark bars present the frequencies of finding the design with minimal total sample size, N_{max}, in the 100 replications. The light bars present the frequency of finding minimax design with minimal $E(N \mid H_0)$ among all successes cases of identifying the minimal N_{max}.

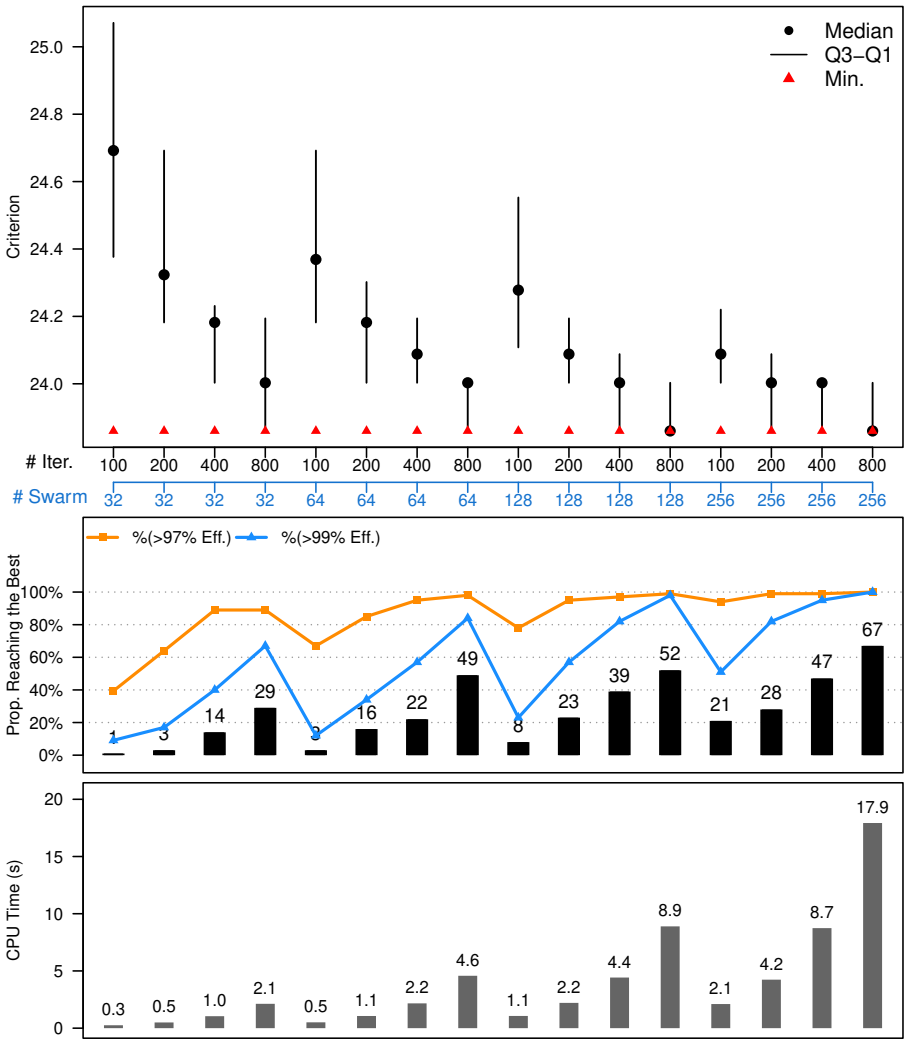


Figure 2: The performances of the three-stage optimal design search PSO algorithm using different configurations of swarm size and number of iterations. For each configuration, we conduct 100 times of replications starting with different initial swarms. For interpretation of this figure, please refer to Figure 1(a).

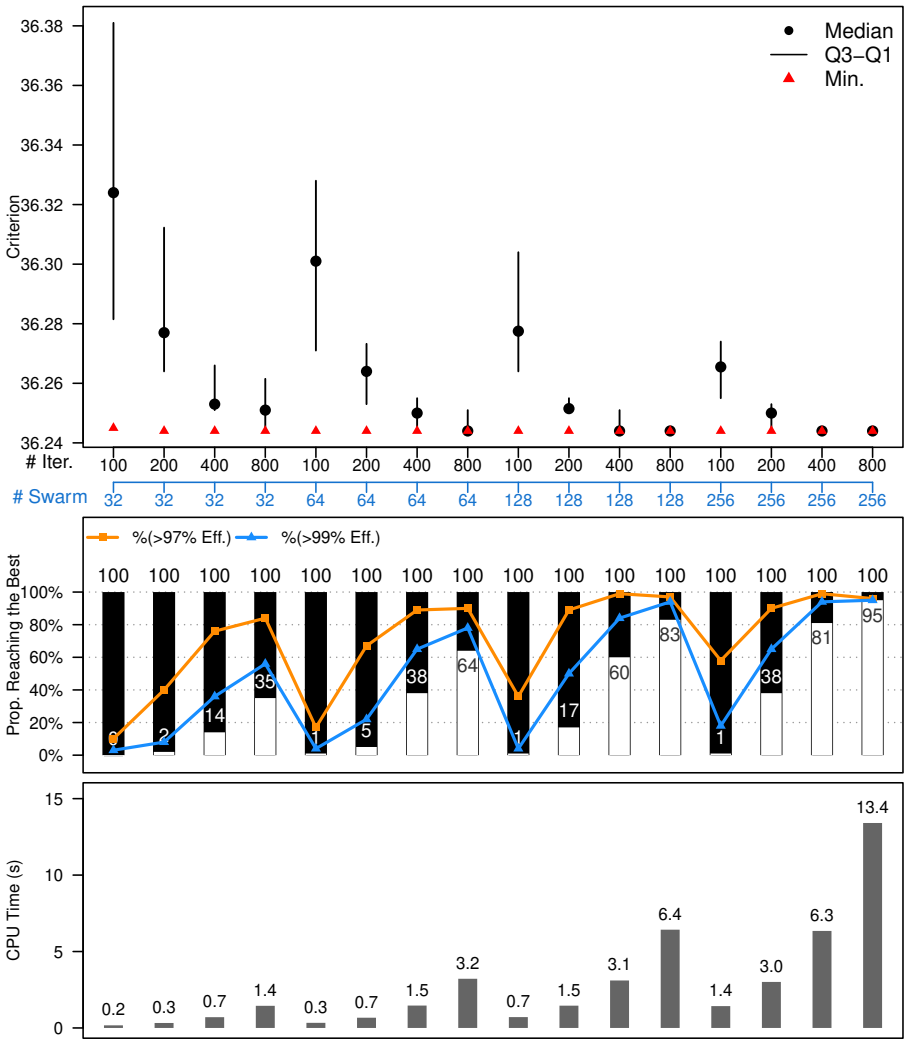


Figure 3: The performances of the three-stage minimax design search PSO algorithm using different configurations of swarm size and number of iterations. For each configuration, we conduct 100 times of replications starting with different initial swarms. For interpretation of this figure, please refer to Figure 1(b).

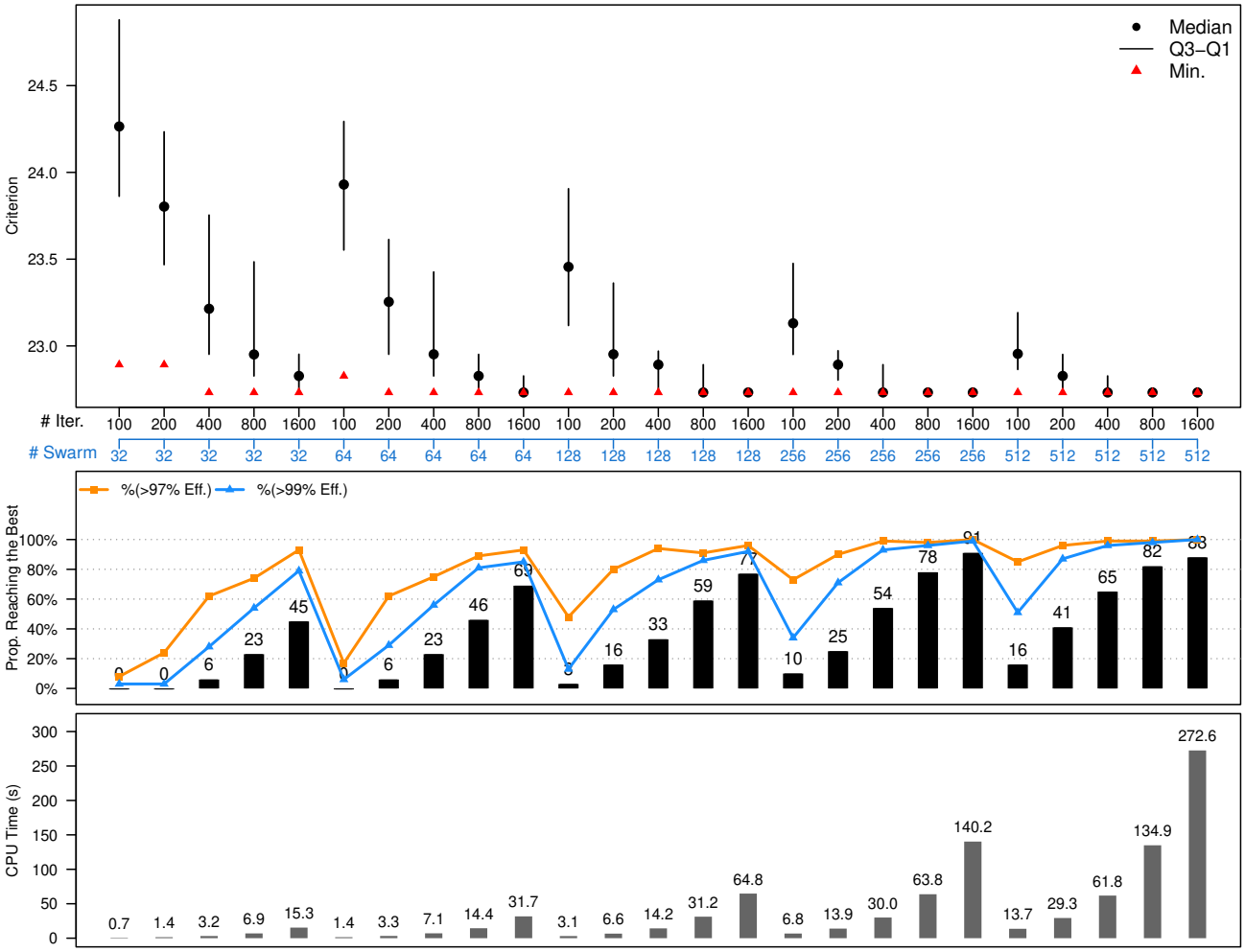


Figure 4: The performances of the four-stage optimal design search PSO algorithm using different configurations of swarm size and number of iterations. For each configuration, we conduct 100 times of replications starting with different initial swarms. For interpretation of this figure, please refer to Figure 1(a).

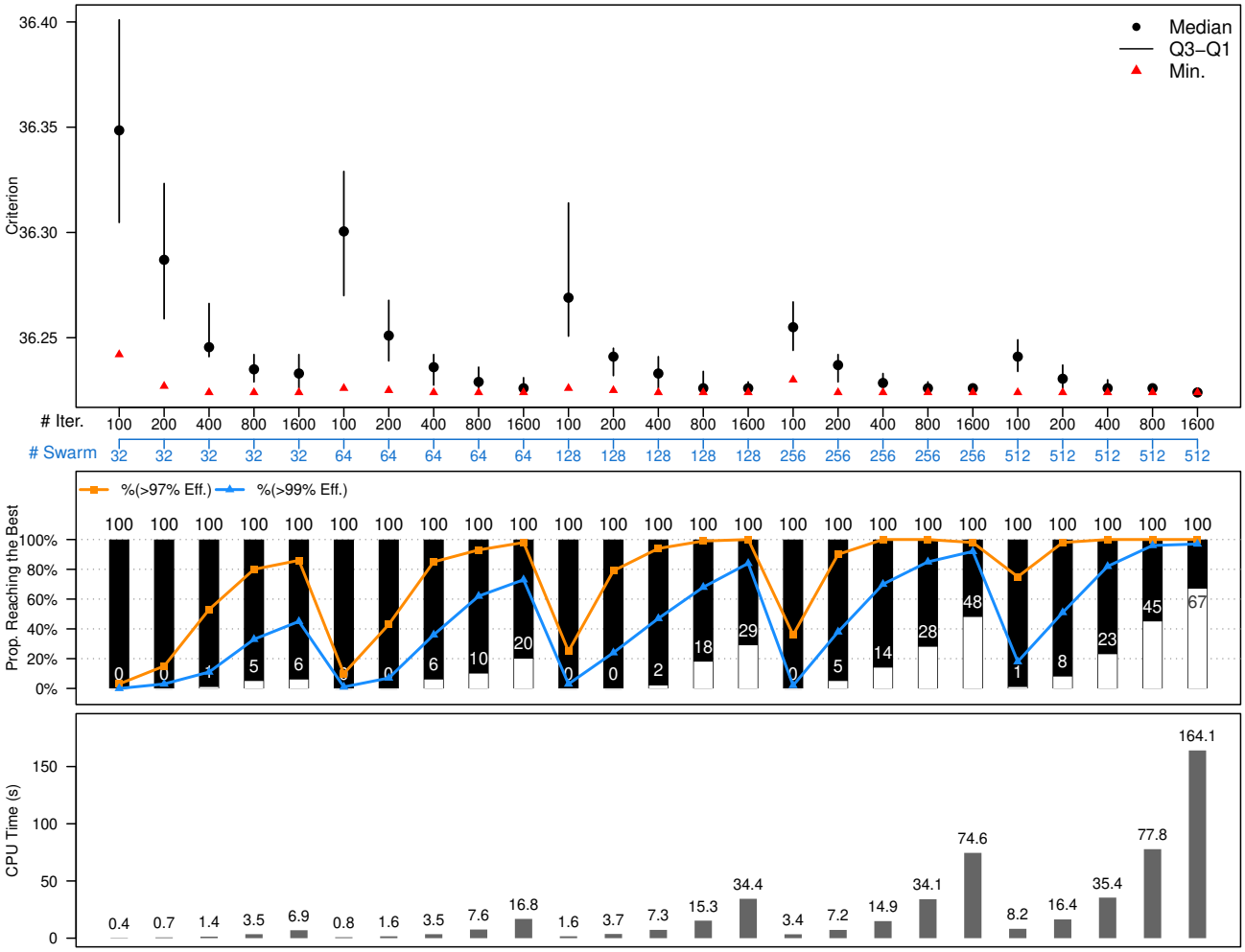


Figure 5: The performances of the four-stage minimax design search PSO algorithm using different configurations of swarm size and number of iterations. For each configuration, we conduct 100 times of replications starting with different initial swarms. For interpretation of this figure, please refer to Figure 1(b).

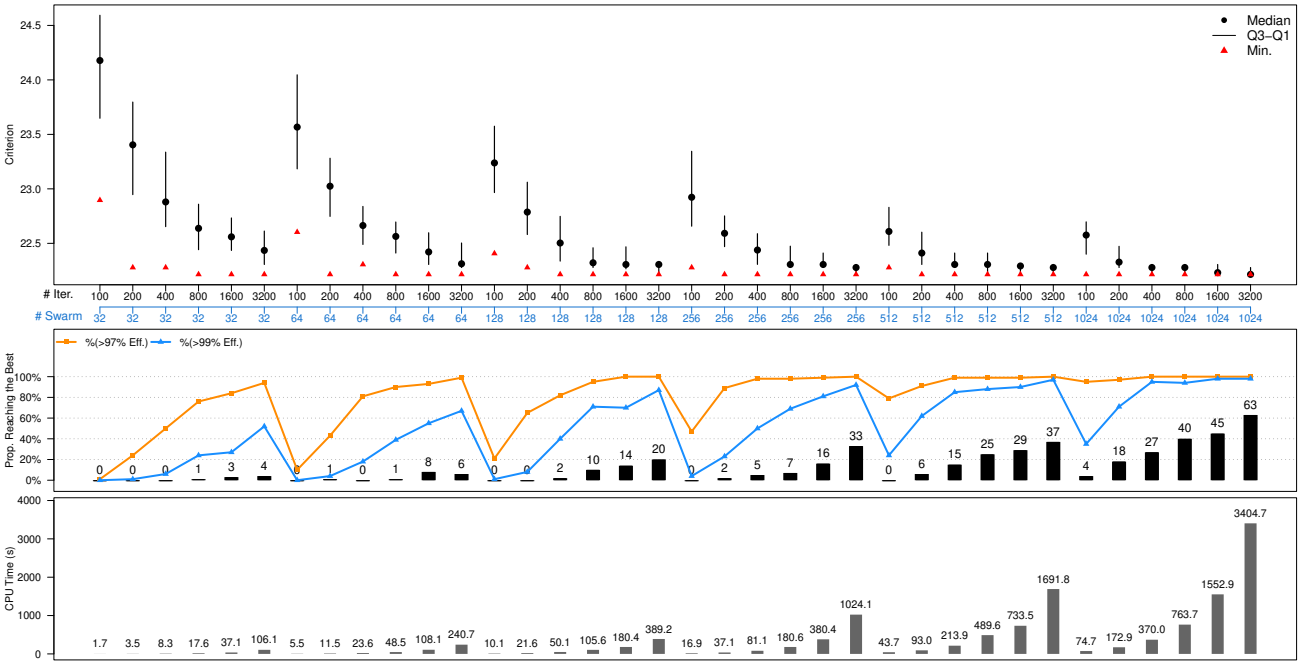


Figure 6: The performances of the five-stage optimal design search PSO algorithm using different configurations of swarm size and number of iterations. For each configuration, we conduct 100 times of replications starting with different initial swarms. For interpretation of this figure, please refer to Figure 1(a).

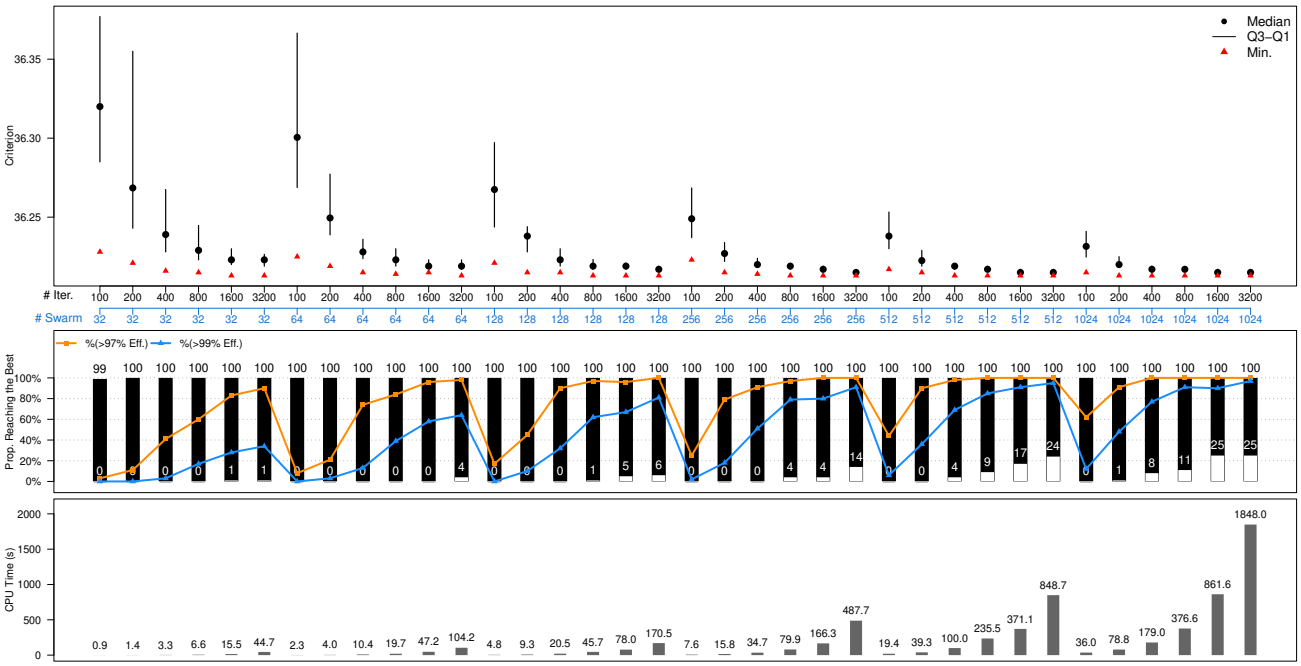


Figure 7: The performances of the five-stage minimax design search PSO algorithm using different configurations of swarm size and number of iterations. For each configuration, we conduct 100 times of replications starting with different initial swarms. For interpretation of this figure, please refer to Figure 1(b).

4.3 Catalogs of the Multi-stage Designs

We show the new results of four- and five-stage designs generated by the proposed algorithm in this paper. Following Section 4.2, to find the best design based on optimal and minimax criteria, the swarm size is set to be 2^{k+5}, and the maximal number of iterations is set to be $100 \times 2^{k-1}$ for $k = 4, 5$.

Tables 8 and 9 show the full catalog of the PSO-generated four-stage designs following the cases of (p_0, p_1) and (α, β) considered in Chen (1997). To save space, we present designs for selected cases when (p_0, p_1) and (α, β) for the five-stage designs.

From the table, we note that, for small p_0, the PSO finds the degenerated four- and five-stage designs suggesting the users to perform clinical trials via three stages. Here, the degenerated design is those designs with the same number for r_k and r_{k+1} for some k, meaning that there is no go/no-go decision to be made at the $(k+1)$th interim look. For example, in Tables 9 and 11, when $(p_0, p_1) = (0.05, 0.25)$ and $(\alpha, \beta) = (0.1, 0.1)$ the four-stage optimal design is $\{0/9, 1/18, 1/19, 2/26\}$, and the five-stage one is $\{0/9, 1/18, 1/24, 1/25, 2/26\}$. Both four- and five-stage optimal designs are of $E(N \mid H_0) = 13.78$ which is identical to the optimal three-stage design, $\{0/9, 1/18, 2/26\}$, in Chen (1997). Interestingly, the resulting four and five-stage optimal designs have identical sample sizes and cutoff boundaries for the first and final stages as in Chen's results.

From the statistical viewpoint, designs with fewer stages appear to be adequate in practice. Our results show that increasing the number of interim looks does not help improve the expected sample size. When the sample size is reduced, we rerun PSO to search for the optimal design. In practice, suppose the number of stages is fixed and the aim is to generate a meaningful multi-stage design that does not degenerate to a design with fewer stages. Recall that in Section 3, we can adopt the constraint on the r_ks by setting $(q_1, q_2, \ldots, q_K) = (0, 1, 1, ..., 1)$ by forcing the cutoff boundary to increase by one patient at every stage. Then PSO finds a four-stage constrained optimal design with $\{0/10, 1/17, 2/24, 3/32\}$ and the corresponding expected sample size $E(N \mid H_0) = 14.78$. This constrained design has 93.23% efficiency relative to the optimal design.

For minimax designs, degenerated designs can also be observed for small p_0. The results of the four and five-stage minimax designs for $(p_0, p_1) = (0.05, 0.25)$ and $(\alpha, \beta) = (0.1, 0.1)$ are presented in Tables 9 and 11, which are $\{0/13, 0/16, 1/18, 2/20\}$ and $\{0/13, 0/15, 0/17, 1/18, 2/20\}$, respectively. Their maximal sample sizes are both 20 patients which is identical to the three-stage minimax design $\{0/13, 1/18, 2/20\}$ found in Chen (1997), and, they have the same expected

sample size $E(N \mid H_0) = 15.86$. Similarly, suppose a four-stage design is sought and the constraint on increasing the r_k's is adopted. The resulting constrained minimax design found by PSO is $\{0/10, 1/17, 2/26, 3/30\}$ which requires 10 more patients than the minimax design does. On the other hand, the expected sample size of the constrained minimax design is $E(N \mid H_0) = 14.82$, which is smaller than that of the minimax design.

Table 8: Catalog of PSO-Generated four-stage Designs ($p_1 - p_0 = 0.15$). For each case of (p_0, p_1), the table shows three resulting designs under error constraints $(\alpha, \beta) = (0.1, 0.1), (0.05, 0.2), (0.05, 0.1)$, respectively.

p_0	p_1	Optimal design	T1E	Pow	$E(N\|H_0)$	Minimax design	T1E	Pow	$E(N\|H_0)$
0.05	0.2	0/13, 1/22, 2/32, 3/38	0.09	0.90	21.00	0/18, 1/26, 2/31, 3/32	0.07	0.90	24.77
		0/10, 1/19, 2/26, 3/31	0.05	0.80	15.55	0/14, 1/20, 2/26, 3/27	0.04	0.80	18.68
		0/14, 1/24, 2/30, 4/43	0.05	0.90	22.98	0/24, 1/30, 2/34, 4/38	0.04	0.90	30.93
0.1	0.25	0/12, 2/23, 4/36, 7/50	0.10	0.90	28.12	0/19, 2/28, 3/33, 6/40	0.10	0.90	32.24
		0/ 8, 2/20, 4/31, 8/53	0.05	0.80	21.01	0/13, 1/19, 3/28, 7/40	0.04	0.80	25.80
		1/17, 3/29, 6/47, 10/67	0.05	0.90	31.26	1/22, 3/33, 5/44, 9/55	0.04	0.90	36.31
0.15	0.3	1/15, 4/28, 7/42, 12/62	0.10	0.90	33.91	2/23, 4/33, 6/40, 11/53	0.09	0.90	38.31
		1/12, 3/21, 7/36, 13/63	0.05	0.80	25.69	1/16, 3/24, 6/36, 11/48	0.04	0.80	30.47
		2/19, 6/37, 10/54, 17/84	0.05	0.90	38.03	3/31, 6/44, 9/55, 14/64	0.05	0.90	47.93
0.2	0.35	2/17, 5/28, 10/46, 17/68	0.10	0.90	38.62	3/24, 7/38, 10/48, 15/58	0.10	0.90	42.33
		1/11, 5/24, 11/45, 18/68	0.05	0.80	29.58	3/22, 6/32, 9/43, 15/53	0.05	0.80	36.55
		3/21, 7/35, 13/55, 24/93	0.05	0.90	43.57	16/65, 18/71, 19/72, 20/74	0.05	0.90	66.07
0.25	0.4	4/21, 9/37, 14/50, 23/77	0.10	0.90	42.36	5/29, 9/40, 14/54, 20/64	0.10	0.90	48.49
		3/15, 7/27, 15/50, 25/80	0.05	0.80	32.40	6/27, 10/38, 14/48, 20/60	0.05	0.80	38.14
		4/21, 11/42, 20/69, 32/103	0.05	0.90	47.90	8/44, 15/63, 20/74, 27/83	0.05	0.90	67.48
0.3	0.45	4/19, 10/34, 17/54, 28/79	0.10	0.90	45.07	8/32, 12/42, 18/56, 25/69	0.10	0.90	48.91
		3/14, 8/26, 17/50, 29/78	0.05	0.80	34.64	8/29, 13/40, 19/55, 25/65	0.05	0.80	40.51
		6/24, 12/39, 23/68, 41/114	0.05	0.90	51.00	15/52, 23/72, 31/85, 33/88	0.05	0.90	66.05
0.35	0.5	5/19, 12/35, 22/58, 35/87	0.10	0.90	46.63	9/33, 16/49, 23/62, 30/72	0.10	0.90	55.48
		4/14, 10/28, 21/53, 38/91	0.05	0.80	36.10	11/36, 15/42, 21/55, 29/66	0.05	0.80	47.51
		8/26, 16/45, 27/69, 46/111	0.05	0.90	52.87	11/36, 19/54, 28/74, 40/94	0.05	0.90	60.24
0.4	0.55	6/19, 12/32, 23/54, 40/88	0.10	0.90	47.62	9/28, 16/43, 23/56, 34/73	0.10	0.90	52.26
		5/15, 12/29, 23/51, 43/92	0.05	0.80	36.55	12/32, 19/44, 31/65, 34/70	0.05	0.80	44.45
		10/27, 21/50, 37/83, 55/119	0.05	0.90	53.76	14/44, 23/61, 33/79, 45/94	0.05	0.90	73.36
0.45	0.6	9/23, 17/38, 29/60, 43/85	0.10	0.90	47.01	18/41, 35/69, 36/72, 38/74	0.10	0.90	55.43
		7/17, 14/30, 26/53, 45/86	0.05	0.80	36.50	20/45, 34/64, 36/67, 37/69	0.05	0.80	54.23
		10/25, 22/47, 38/76, 61/119	0.05	0.90	53.48	24/55, 32/69, 47/90, 50/95	0.05	0.90	69.92
0.5	0.65	8/19, 16/33, 29/55, 48/86	0.09	0.90	46.33	13/31, 21/44, 29/57, 41/72	0.10	0.90	53.52
		6/14, 14/27, 28/51, 53/93	0.05	0.80	35.86	17/35, 26/48, 38/65, 40/68	0.05	0.80	45.57
		11/24, 22/43, 38/70, 66/117	0.05	0.90	51.91	19/43, 31/63, 42/79, 54/93	0.05	0.90	68.69
0.55	0.7	9/19, 17/32, 31/53, 48/79	0.10	0.90	43.59	23/43, 31/55, 36/60, 42/68	0.10	0.90	52.42
		7/14, 16/28, 30/50, 55/89	0.05	0.80	33.85	12/24, 25/44, 41/64, 42/66	0.05	0.80	42.86
		11/22, 22/39, 41/69, 70/114	0.05	0.90	49.45	28/51, 41/70, 52/84, 56/89	0.05	0.90	63.11
0.6	0.75	9/17, 17/29, 30/48, 50/76	0.10	0.90	40.34	16/31, 24/42, 35/56, 43/64	0.10	0.90	50.06
		7/13, 16/26, 31/47, 53/79	0.05	0.80	31.40	16/28, 28/45, 40/58, 42/61	0.05	0.80	41.48
		9/17, 22/36, 41/64, 70/105	0.05	0.90	45.88	28/46, 39/63, 50/75, 57/84	0.05	0.90	56.96
0.65	0.8	10/17, 21/32, 34/49, 48/68	0.10	0.90	36.02	17/28, 25/38, 41/57, 42/59	0.10	0.90	41.70
		8/13, 19/28, 34/48, 50/69	0.05	0.80	28.19	12/21, 23/35, 33/47, 41/55	0.05	0.80	36.95
		10/17, 22/33, 39/56, 70/98	0.05	0.90	40.99	21/36, 32/50, 44/64, 55/75	0.05	0.90	55.88
0.7	0.85	8/13, 20/28, 31/42, 44/58	0.10	0.90	30.71	15/23, 26/36, 36/47, 39/51	0.10	0.90	34.98
		5/ 8, 15/21, 27/36, 51/66	0.05	0.80	24.23	10/16, 18/26, 25/34, 39/49	0.05	0.80	29.81
		9/14, 23/32, 38/51, 63/82	0.05	0.90	35.07	18/27, 31/43, 50/64, 52/67	0.05	0.90	42.89
0.75	0.9	5/ 8, 15/20, 25/32, 39/48	0.09	0.90	24.29	11/17, 19/26, 25/32, 33/40	0.10	0.90	29.13
		3/ 5, 14/18, 26/32, 42/51	0.05	0.80	19.22	8/12, 16/21, 25/31, 33/39	0.05	0.80	22.66
		10/14, 20/26, 32/40, 51/62	0.05	0.90	27.68	10/15, 18/24, 29/37, 45/54	0.05	0.90	30.38
0.8	0.95	6/ 8, 14/17, 21/25, 30/35	0.10	0.90	16.43	1/ 3, 10/13, 17/20, 26/30	0.10	0.90	17.44
		1/ 2, 8/10, 18/21, 35/40	0.05	0.80	13.24	3/ 5, 7/ 9, 16/19, 26/29	0.05	0.80	14.36
		5/ 7, 13/16, 23/27, 41/47	0.05	0.90	18.64	12/16, 20/24, 31/35, 35/40	0.05	0.90	23.99

Table 9: Catalog of PSO-Generated four-stage Designs ($p_1 - p_0 = 0.20$). For each case of (p_0, p_1), the table shows three resulting designs under error constraints $(\alpha, \beta) = (0.1, 0.1), (0.05, 0.2), (0.05, 0.1)$, respectively.

p_0	p_1	Optimal design	T1E	Pow	$E(N\|H_0)$	Minimax design	T1E	Pow	$E(N\|H_0)$
0.05	0.25	0/ 9, 1/18, 1/19, 2/26	0.10	0.90	13.78	0/13, 0/16, 1/18, 2/20	0.07	0.90	15.86
		0/ 8, 1/13, 1/15, 2/19	0.05	0.80	10.41	0/12, 0/14, 1/15, 2/16	0.04	0.80	13.55
		0/10, 1/17, 2/24, 3/32	0.05	0.90	14.78	0/15, 1/21, 1/23, 3/25	0.03	0.90	19.29
0.1	0.3	0/11, 1/15, 2/19, 4/26	0.10	0.90	17.37	0/11, 1/18, 2/22, 4/25	0.09	0.90	18.84
		0/ 7, 1/11, 2/16, 5/30	0.05	0.80	12.79	0/11, 1/16, 2/20, 5/25	0.03	0.80	17.74
		0/ 9, 2/18, 4/30, 7/45	0.05	0.90	19.37	1/18, 2/23, 3/27, 6/33	0.04	0.90	23.82
0.15	0.35	0/ 9, 2/16, 4/26, 7/33	0.10	0.90	20.55	0/11, 2/17, 4/25, 7/32	0.09	0.90	21.75
		0/ 6, 2/13, 4/23, 8/35	0.05	0.80	15.25	1/12, 2/16, 4/23, 7/28	0.05	0.80	18.16
		1/12, 3/20, 6/31, 11/51	0.05	0.90	22.79	1/17, 3/24, 6/33, 9/38	0.05	0.90	27.59
0.2	0.4	1/10, 4/21, 7/30, 11/43	0.10	0.90	22.73	1/13, 3/20, 6/29, 10/36	0.08	0.90	25.70
		1/ 8, 4/18, 7/28, 12/43	0.05	0.80	17.26	1/11, 3/16, 6/25, 10/33	0.05	0.80	19.40
		2/14, 5/23, 9/37, 15/55	0.05	0.90	25.52	2/16, 5/25, 9/39, 13/45	0.05	0.90	28.11
0.25	0.45	1/10, 4/18, 8/28, 14/45	0.10	0.90	24.42	5/23, 9/31, 11/35, 12/37	0.10	0.90	28.41
		1/ 8, 3/13, 7/23, 15/45	0.05	0.80	18.65	2/13, 5/20, 8/29, 13/36	0.04	0.80	22.63
		2/13, 5/21, 11/36, 19/58	0.05	0.90	27.68	3/19, 7/30, 11/39, 17/49	0.04	0.90	33.68
0.3	0.5	2/11, 6/21, 11/33, 17/46	0.10	0.90	25.77	3/20, 8/30, 12/36, 15/39	0.09	0.90	33.10
		2/ 9, 5/16, 11/30, 20/53	0.05	0.80	19.79	3/15, 12/31, 14/35, 15/37	0.05	0.80	26.80
		2/11, 6/21, 13/38, 23/60	0.05	0.90	29.05	5/20, 9/30, 14/41, 21/53	0.05	0.90	32.36
0.35	0.55	3/13, 7/21, 12/31, 19/45	0.10	0.90	26.51	3/14, 7/21, 11/30, 18/42	0.10	0.90	27.17
		2/ 8, 5/15, 11/27, 22/50	0.05	0.80	20.26	3/12, 6/18, 10/26, 18/39	0.05	0.80	22.50
		3/12, 10/27, 17/42, 28/65	0.05	0.90	29.71	8/30, 13/39, 19/48, 24/53	0.04	0.90	42.56
0.4	0.6	3/11, 8/20, 14/33, 23/49	0.10	0.90	26.29	6/19, 11/29, 16/37, 20/41	0.10	0.90	30.82
		2/ 7, 7/17, 14/31, 25/51	0.05	0.80	20.27	5/15, 9/22, 16/33, 20/39	0.05	0.80	23.76
		5/15, 11/26, 20/43, 33/69	0.05	0.90	29.98	6/19, 10/26, 17/39, 27/54	0.05	0.90	33.47
0.45	0.65	3/10, 9/20, 16/33, 25/48	0.10	0.90	25.81	5/15, 8/20, 14/30, 22/41	0.10	0.90	27.98
		2/ 7, 7/15, 15/29, 27/50	0.05	0.80	19.97	5/13, 11/24, 15/29, 22/39	0.05	0.80	22.67
		5/13, 12/26, 21/41, 37/70	0.05	0.90	29.43	9/23, 16/34, 28/51, 29/53	0.05	0.90	35.69
0.5	0.7	3/ 9, 8/17, 16/30, 26/45	0.10	0.90	24.89	5/14, 10/22, 17/32, 23/39	0.10	0.90	27.93
		4/ 9, 9/17, 15/27, 29/49	0.05	0.80	19.23	5/13, 11/22, 15/28, 23/37	0.05	0.80	24.11
		5/12, 12/23, 22/40, 38/65	0.05	0.90	28.12	10/22, 19/35, 30/50, 31/52	0.05	0.90	33.32
0.55	0.75	5/11, 10/18, 19/32, 28/45	0.10	0.90	23.20	6/13, 13/24, 22/35, 24/38	0.10	0.90	25.00
		3/ 7, 8/14, 16/26, 29/45	0.05	0.80	17.79	6/12, 9/17, 16/26, 24/36	0.05	0.80	20.09
		5/11, 11/20, 21/35, 36/56	0.05	0.90	26.26	9/18, 16/28, 23/38, 32/49	0.05	0.90	28.90
0.6	0.8	3/ 7, 8/14, 18/28, 27/40	0.10	0.90	20.90	7/13, 11/19, 16/25, 24/35	0.10	0.90	21.56
		3/ 6, 10/16, 18/27, 28/40	0.05	0.80	16.23	6/11, 15/23, 23/32, 24/34	0.05	0.80	19.44
		5/10, 13/21, 21/32, 36/52	0.05	0.90	23.88	11/21, 17/29, 24/37, 32/45	0.04	0.90	32.06
0.65	0.85	5/ 9, 9/14, 17/24, 26/36	0.10	0.90	18.21	4/ 8, 11/17, 18/26, 23/31	0.09	0.90	19.03
		2/ 4, 7/11, 15/21, 29/39	0.05	0.80	14.18	5/ 9, 8/12, 16/22, 23/30	0.05	0.80	15.41
		5/ 9, 12/18, 20/28, 35/47	0.05	0.90	20.67	12/19, 20/28, 27/36, 30/40	0.05	0.90	25.03
0.7	0.9	2/ 4, 8/12, 15/20, 24/31	0.10	0.90	14.79	5/ 9, 11/16, 15/20, 20/25	0.09	0.90	17.02
		4/ 6, 9/12, 17/22, 24/30	0.05	0.80	11.53	4/ 7, 11/15, 19/23, 20/25	0.05	0.80	14.55
		5/ 8, 10/14, 16/21, 31/39	0.05	0.90	16.64	3/ 7, 11/16, 13/18, 26/32	0.05	0.90	20.39
0.75	0.95	3/ 5, 6/ 8, 11/14, 16/19	0.10	0.90	10.23	3/ 5, 6/ 8, 11/14, 16/19	0.10	0.90	10.23
		1/ 2, 5/ 7, 9/11, 19/22	0.04	0.80	8.10	4/ 6, 8/10, 14/16, 17/20	0.05	0.80	9.85
		3/ 5, 6/ 8, 13/16, 24/28	0.04	0.90	11.83	6/ 9, 11/14, 19/22, 22/26	0.05	0.90	14.49

Table 10: Catalog of PSO-Generated five-stage Designs ($p_1 - p_0 = 0.15$). For each case of (p_0, p_1), the table shows the resulting designs under error constraint $(\alpha, \beta) = (0.1, 0.1)$, respectively.

p_0	p_1	Optimal design	T1E	Pow	$E(N\|H_0)$	Minimax design	T1E	Pow	$E(N\|H_0)$
0.05	0.2	0/13, 0/21, 1/22, 2/32, 3/38	0.09	0.90	21.00	0/18, 1/26, 1/30, 2/31, 3/32	0.07	0.90	24.77
0.1	0.25	0/12, 1/20, 2/23, 4/35, 7/51	0.10	0.90	27.46	0/19, 1/24, 2/29, 3/33, 6/40	0.10	0.90	31.77
0.2	0.35	2/18, 4/25, 8/38, 12/52, 17/69	0.10	0.90	37.81	3/24, 5/32, 7/39, 10/48, 15/58	0.10	0.90	41.76
0.3	0.45	4/19, 9/32, 14/46, 19/59, 28/79	0.10	0.90	44.09	9/34, 14/48, 17/54, 20/61, 25/69	0.10	0.90	49.00
0.4	0.55	6/19, 14/36, 20/48, 29/67, 41/91	0.10	0.90	46.33	13/36, 20/50, 26/63, 29/67, 34/73	0.10	0.90	52.59
0.5	0.65	9/21, 16/33, 25/48, 37/68, 49/89	0.10	0.90	45.10	17/37, 18/40, 27/54, 36/67, 41/72	0.10	0.90	53.97
0.7	0.85	8/13, 15/22, 21/30, 28/38, 44/58	0.10	0.90	29.74	2/ 6, 2/ 8, 25/35, 36/47, 39/51	0.10	0.90	37.74
0.8	0.95	4/ 6, 6/ 8, 14/17, 21/25, 30/35	0.10	0.90	15.74	4/ 6, 9/12, 13/16, 21/25, 27/31	0.09	0.90	15.97

Table 11: Catalog of PSO-Generated five-stage Designs ($p_1 - p_0 = 0.20$). For each case of (p_0, p_1), the table shows the resulting designs under error constraint $(\alpha, \beta) = (0.1, 0.1)$, respectively.

p_0	p_1	Optimal design	T1E	Pow	$E(N\|H_0)$	Minimax design	T1E	Pow	$E(N\|H_0)$
0.05	0.25	0/ 9, 1/18, 1/24, 1/25, 2/26	0.10	0.90	13.78	0/13, 0/15, 0/17, 1/18, 2/20	0.07	0.90	15.86
0.1	0.3	0/10, 1/16, 2/20, 3/24, 4/26	0.10	0.90	17.21	0/11, 1/18, 2/22, 3/24, 4/25	0.09	0.90	18.70
0.2	0.4	1/11, 3/18, 5/25, 7/30, 10/38	0.10	0.90	22.21	1/13, 2/18, 4/23, 6/29, 10/36	0.08	0.90	25.30
0.3	0.5	2/11, 5/18, 8/26, 13/38, 18/50	0.10	0.90	25.10	4/22, 7/29, 9/32, 12/36, 15/39	0.09	0.90	32.62
0.4	0.6	3/11, 6/17, 9/23, 14/32, 23/49	0.10	0.90	25.64	4/16, 8/23, 11/29, 16/37, 20/41	0.10	0.90	30.28
0.5	0.7	3/ 9, 8/17, 13/25, 20/37, 27/47	0.09	0.90	24.29	5/14, 10/22, 14/28, 19/35, 23/39	0.10	0.90	27.40
0.6	0.8	3/ 7, 8/14, 15/24, 19/30, 27/40	0.10	0.90	20.23	7/13, 20/30, 22/33, 23/34, 24/35	0.10	0.90	23.51
0.7	0.9	4/ 7, 8/12, 11/15, 17/22, 21/27	0.10	0.90	14.30	5/ 9, 9/14, 12/17, 16/21, 20/25	0.09	0.90	16.61

4.4 Insights of Multi-stage Designs and Number of Stages

In the previous sections, we compared Simon's designs with different number of stages by focusing on the numbers $E(N \mid H_0)$ and N_{max}. To fix ideas, we consider eight scenarios of (p_0, p_1) when $(\alpha, \beta) = (0.1, 0.1)$. The designs for comparison come from Simon (1989) and Chen (1997),and they include the four- and five-stage designs, and the known two- and three-stage designs.

Figures 8 and 9 present the values of $E(N \mid H_0)$ and N_{max} for the optimal and minimax designs, respectively, from two to five-stages. Each subplot represents one of the eight different (p_0, p_1) settings. The height of each bar indicates the required sample size, the blue bars represent $E(N \mid H_0)$ and the orange bars represent N_{max}. The comparison between the two bar colors highlights the extent to which the trial might require fewer participants in practice, $E(N \mid H_0)$, versus the maximum number of participants, N_{max}, if the trial reaches its full extent without early termination.

In general, a larger number of stages tends to reduce the expected sample size required to complete the trial. However, N_{max} exhibits a different pattern. The maximal sample size remains relatively stable across stages in most scenarios. Particularly for the minimax designs (Figure 9), the maximal sample size is constant across stages except for the scenario of $(p_0, p_1) = (0.70, 0.85)$, indicating that the maximal number of participants required does not benefit as much from the addition of more stages.

Across all scenarios, as the number of stages increases from two to five, the $E(N \mid H_0)$ consistently decreases, demonstrating that multi-stage designs can significantly reduce the average number of participants required. This finding aligns with the principle of interim analyses, where trials are reviewed at multiple points, allowing for early stopping if sufficient evidence is collected. However, N_{max} generally remains stable, indicating that the upper limit of participants does not decrease substantially even with additional stages.

Focusing first on the results within a single row of each figure, the required sample size is strongly influenced by the response rates associated with p_0 and p_1. A key observation is that trials tend to require larger sample sizes when the response rates are at middle levels for both p_0 and p_1, whereas fewer participants are needed when the response rates are either low or high.

For middle-level response rates (e.g., $(p_0, p_1) = (0.30, 0.45)$ or $(p_0, p_1) = (0.50, 0.65)$), the trial requires more participants. This is due to the uncertainty associated with moderate response rates, where it is more difficult to confidently determine whether a treatment is effective. Therefore, the trial design requires more patients to ensure that the study has sufficient power to draw reliable conclusions in these scenarios. In contrast, trials with low or high response rates

are more straightforward in detecting the treatment's effect, allowing for smaller sample sizes.

The upper and lower rows of each figure show results for scenarios with different gaps between p_0 and p_1. The size of the gap between p_0 and p_1 is directly related to the statistical power of the clinical trial, which in turn impacts the required sample sizes. A larger gap between p_0 and p_1 generally leads to higher statistical power, as it becomes easier to detect a significant difference between the two hypotheses, allowing the trial to require fewer participants. Conversely, a smaller gap reduces the power, requiring more participants to detect the effect. In the upper row, where the gap between p_0 and p_1 is consistently smaller (0.15), the statistical power is lower, and the trials need larger values of $E(N \mid H_0)$ and N_{max} across all stages. On the other hand, in the lower row, where the gap between p_0 and p_1 is larger (0.20), the trials benefit from increased statistical power, which translates into smaller sample size requirements.

Figure 10 shows the sample size at each interim look of the K-stage optimal design, $K = 2, 3, 4, 5$.

- The sample size needed for the first stage, n_1, decreases along with the increasing number of stages.

Figure 11 shows the sample size at each interim look of the K-stage minimax design, $K = 2, 3, 4, 5$.

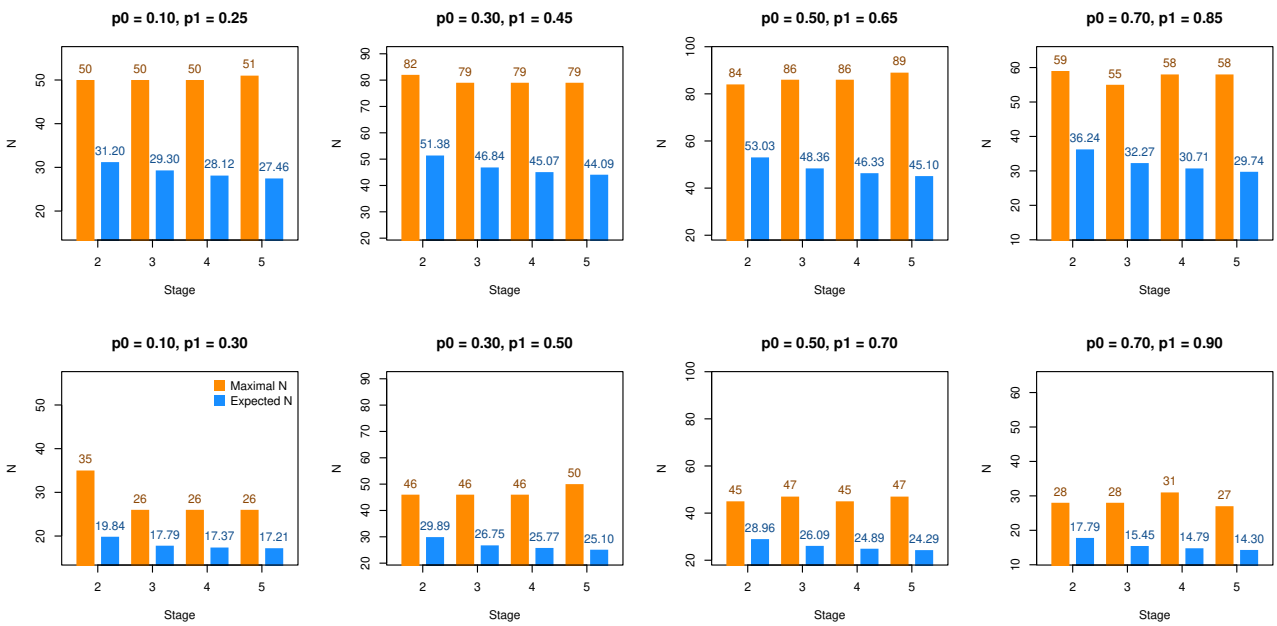


Figure 8: Expected sample size and Maximal sample size of two- to five-stage optimal designs. Eight cases of (p_0, p_1) are selected and (α, β) is fixed as $(0.1, 0.1)$.

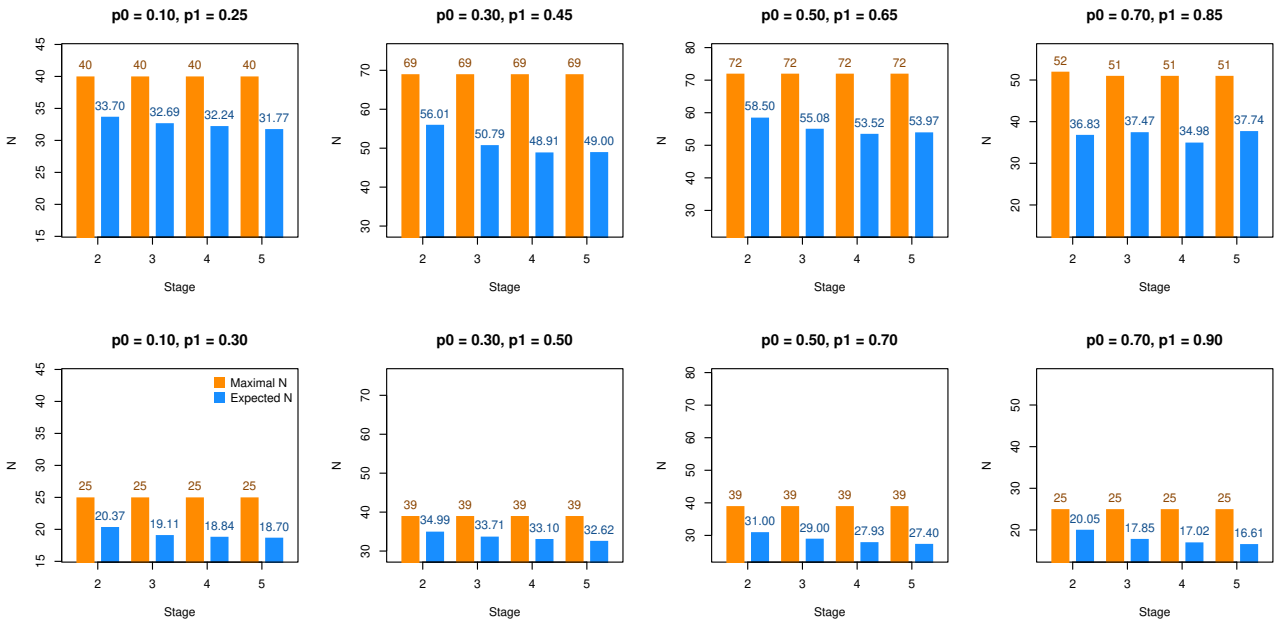


Figure 9: Expected sample size and Maximal sample size of two- to five-stage minimax designs. Eight cases of (p_0, p_1) are selected and (α, β) is fixed as $(0.1, 0.1)$.

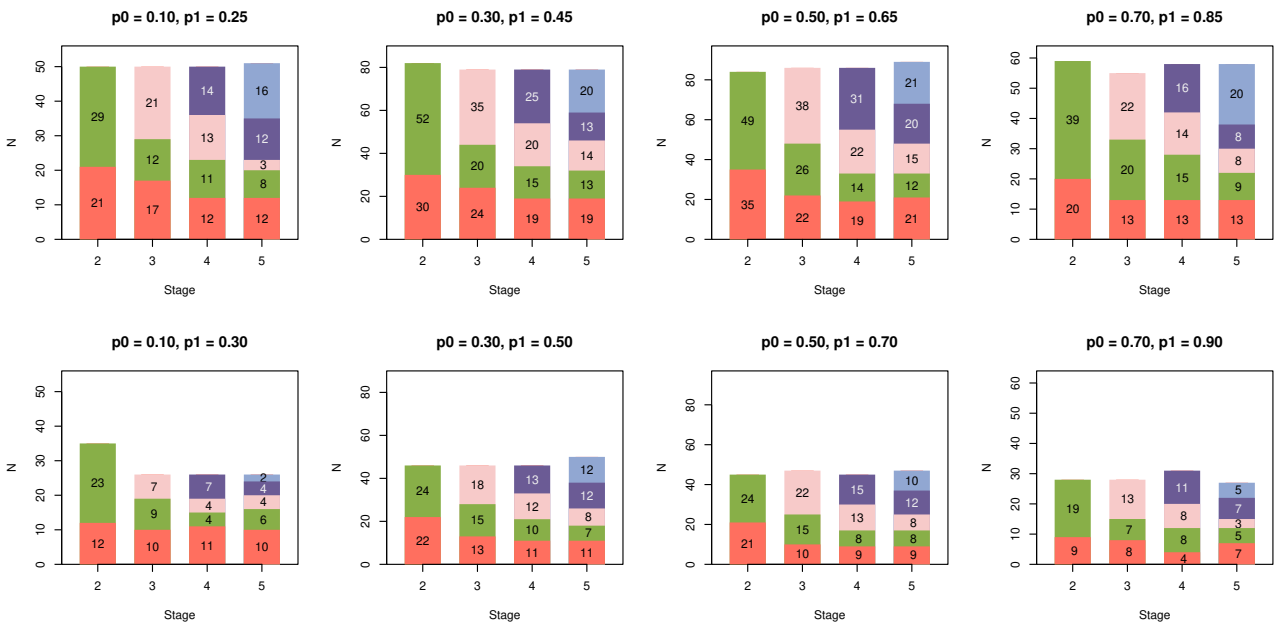


Figure 10: Sample size at each interim look of two- to five-stage optimal designs. Eight cases of (p_0, p_1) are selected and (α, β) is fixed as $(0.1, 0.1)$.

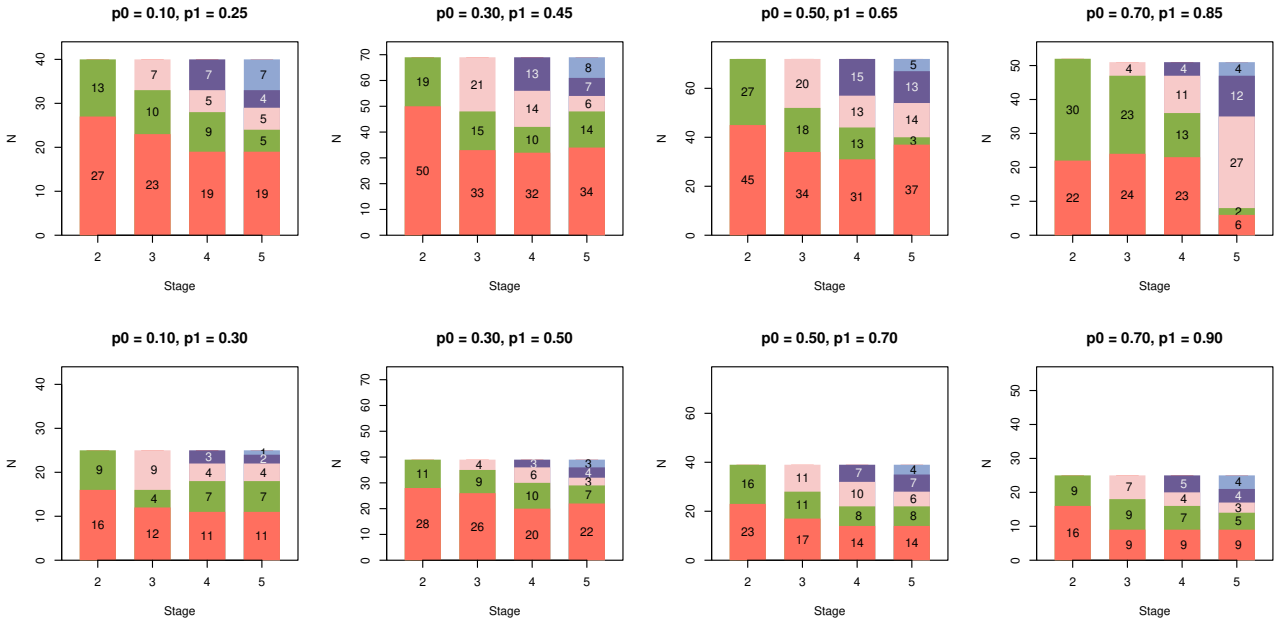


Figure 11: Sample size at each interim look of two- to five-stage minimax designs. Eight cases of (p_0, p_1) are selected and (α, β) is fixed as $(0.1, 0.1)$.

5 Conclusion

We applied particle swarm optimization to tackle challenges in the design of multi-stage trials. There are many different types of variables to optimize and the problem has multiple constraints. In the literature, there is some work on finding optimal two-stage designs, but finding three, four or five-stage Phase II designs are scarce and those who attempted to do so found computational issues were a big challenge. Our work uses a new class of algorithms that are already widely used across disciplines to solve the complex design optimization problem with three to five-stages with multiple constraints.

Our algorithms are PSO-based and codes are available from the first author. Alternatively, the weblink https://github.com/PingYangChen/kStageSimon which contains many codes that were used to find optimal designs in this paper. It should be noted that computing time is still long for finding a five-stage clinical trial, suggesting that further work to enhance the algorithm is required for practical application. A common way to improve performance of a metaheuristic algorithm is to hybridize it with one or more cleverly chosen algorithms, and they may be metaheuristic or a mathematical programming method, such as definite programming method. The guiding principle for hybridization is that the hybridized algorithm should outperform all the algorithms that were used to form it; see Blum and Raidl (2016); Singh and Singh (2017) for details on hybridization techniques.

We conclude with a reminder that PSO is assumptions free and is a general optimization tool. In particular, it can be directly applied to find different types of two-stage adaptive designs such as those described in Shan et al. (2013) and designs in its references. The problems we solved in this paper are nontrivial and the solutions are useful. We believe PSO can be used to find harder-to-find optimal multi-stage optimal designs for different scenarios, such as for determining the optimal number of interim looks and when the interim looks should be, the number of patients at each each look, or an allocation scheme to maximize the power of the test, including the case when there are multiple outcomes. Naturally, PSO can be used to tackle optimization problems whenever they arise; Cui et al. (2024) demonstrated its utility for finding various types of estimates for the model parameters and for solving various optimization problems across disciplines.

The website https://biostatistics.mdanderson.org/SoftwareDownload/ at the Biostatistics Department in The University of Texas MD Anderson contains a wealth of useful design and analysis tools for clinical trials. Many of them are Bayesian in nature and all are available for free download. We believe PSO can enhance some of the algorithms by making them more flexible.

For instance, some software finds the multistage designs without a firm optimality criterion and some require the user to specify the number of interim analysis and the time for each of these interim looks before a multistage design is found. Our immediate future work is to use PSO to further enhance some of their multistage designs by making them even more versatile for practitioners.

6 Acknowledgement

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References

Alira Health (2024). Calculator for simon's two-stage design and its modifications. Accessed: 2024-08-01.

Blum, C. and Raidl, G. R. (2016). *Hybrid Metaheuristics: Powerful Tools for Optimization*. Springer International Publishing.

Cabarrou, B., Leconte, E., Sfumato, P., Boher, J., and Filleron, T. (2022). A stratified adaptive two-stage design with co-primary endpoints for phase ii clinical oncology trials. *BMC Medical Research Methodology*, 22:278.

Chen, P.-Y., Chen, R.-B., Tung, H.-C., and Wong, W. K. (2017). Standardized maximim d-optimal designs for enzyme kinetic inhibition models. *Chemometrics and Intelligent Laboratory Systems*, 169:79–86.

Chen, R.-B., Chang, S.-P., Wang, W., Tung, H.-C., and Wong, W. K. (2015). Minimax optimal designs via particle swarm optimization methods. *Statistics and Computing*, 25:975–988.

Chen, R.-B., Chen, P.-Y., Hsu, C.-L., and Wong, W. K. (2020). Hybrid algorithms for generating optimal designs for discriminating multiple nonlinear models under various error distributional assumptions. *PloS one*, 15(10):e0239864.

Chen, T. T. (1997). Optimal three-stage designs for phase II cancer clinical trials. *Statistics in medicine*, 16(23):2701–2711.

Clinical Biostats (2023). Simon's two stage design calculator. Accessed: 2024-08-01.

Cui, E., Zhang, Z., Chen, C., and Wong, W. K. (2024). Applications of nature-inspired metaheuristic algorithms for tackling optimization problems across disciplines. *Scientific Reports*, 14(9403).

Desale, S., Rasool, A., Andhale, S., and Rane, P. (2015). Heuristic and meta-heuristic algorithms and their relevance to the real world: A survey. *International Journal of Computer Engineering in Research Trends*, 2(5):296–304.

Eberhart, R. and Kennedy, J. (1995). A new optimizer using particle swarm theory. In *MHS'95. Proceedings of the sixth international symposium on micro machine and human science*, pages 39–43. Ieee.

Eberhart, R. C. and Shi, Y. (2000). Comparing inertia weights and constriction factors in particle swarm optimization. In *Proceedings of the 2000 congress on evolutionary computation. CEC00 (Cat. No. 00TH8512)*, volume 1, pages 84–88. IEEE.

Ensign, L. G., Gehan, E. A., Kamen, D. S., and Thall, P. F. (1994). An optimal three-stage design for phase i1 clinical trials. *Statistics in Medicine*, 13:1727–1736.

Gogna, A. and Tayal, A. (2013). Metaheuristics: review and application. *Journal of Experimental Theoretical Artificial Intelligence*, 25(4):503–526.

Goll, A. and Bauer, P. (2007). Two-stage designs applying methods differing in costs. *Bioinformatics*, 23(12):1519–1526.

Hanfelt, J. J. (1999). Optimal multi-stage designs for a phase II trial that permits one dose escalation. *Statistics in medicine*, 18(11):1323–1339.

Ivanova, A. (2012). Simon's two-stage design tool. Accessed: 2024-06-01.

Jing, N., Liu, F., Wu, C., Zhou, H., and Chen, C. (2022). An optimal two-stage exploratory basket trial design with aggregated futility analysis. *Contemporary Clinical Trials*, 116:106741.

Jung, S. H. (2015). Statistical issues for design and analysis of single-arm multi-stage phase ii cancer clinical trials. *Contemporary Clinical Trials*, 42:9–17.

Kim, J. and Schell, M. J. (2019). Modified simon's minimax and optimal two-stage designs for single-arm phase ii cancer clinical trials. *Oncotarget*, 10(42):4255–4261.

Kim, S. and Wong, W. K. (2018). Extended two-stage adaptive designs with three target responses for phase ii clinical trials. *Statistical methods in medical research*, 27(12):3628–3642.

Korani, W. and Mouhoub, M. (2021). Review on nature-inspired algorithms. *SN Operations Research Forum*, 2(36):https://doi.org/10.1007/s43069–021–00068–x.

Lin, Y. and Shih, W. J. (2004). Adaptive two-stage designs for single-arm phase IIa cancer clinical trials. *Biometrics*, 60(2):482–490.

Liu, B. (2022). *A Single-Arm Multi-Stage Design & a Flexible Pick-The-Winner Design for Phase II Clinical Trials*. PhD thesis, University of Kansas.

Liu, X., Yue, R. X., Zhang, Z., and Wong, W. K. (2021). G-optimal designs for hierarchical linear models: an equivalence theorem and a nature-inspired meta-heuristic algorithm. *Soft Computing*, 25:13549–13565.

NCSS Software (2024). *PASS: Two-Stage Designs for Tests of One Proportion (Simon)*. Accessed: 2024-08-01.

Oliveira, H. A. J. (2015). *Overview of Current Metaheuristic Paradigms*. Springer.

Pilz, M., Kunzmann, K., Herrmann, C., Rauch, G., and Kieser, M. (2021). Optimal planning of adaptive two-stage designs. *Statistics in Medicine*, 40:3196–3213.

Poli, R., Kennedy, J., and Blackwell, T. (2007). Particle Swarm Optimization: An Overview. *Swarm Intell*, 1:33–57.

Qiu, J., Chen, R.-B., Wang, W., and Wong, W. K. (2014). Using animal instincts to design efficient biomedical studies via particle swarm optimization. *Swarm and evolutionary computation*, 18:1–10.

Schultz, J., Nichol, F., Elfring, G., and Weed, S. (1973a). Multiple-stage procedures for drug screening. *Biometrics*, pages 293–300.

Schultz, J. R., Nichol, F. R., Elfring, G. L., and Weeds, S. D. (1973b). Multiple-stage procedures for drug screening. *Biometrics*, 29:293–300.

Sechan, V. E. and Whiting, K. (2023). *clinfun: Clinical Trial Design and Data Analysis Functions*. R Foundation for Statistical Computing. Accessed: 2024-08-01.

Shan, G., Wilding, G., Hutson, A., and Gerstenberger, S. (2013). Optimal adaptive two-stage designs for early phase ii clinical trials. *Journal of Experimental Theoretical Artificial Intelligence*, 25(4):503–526.

Shan, G., Wilding, G., Hutson, A., and Gerstenberger, S. (2016). Optimal adaptive two-stage designs for early phase ii clinical trials. *Statistics in Medicine*, 35(8):1257–1266.

Siarry, P. E. (2016). *Metaheuristics*. Springer Nature.

Simon, R. (1989). Optimal two-stage designs for phase ii clinical trials. *Controlled clinical trials*, 10(1):1–10.

Singh, N. and Singh, S. B. (2017). Hybrid algorithm of particle swarm optimization and grey wolf optimizer for improving convergence performance. *Journal of Applied Mathematics*, 606:Article ID 2030489.

Slob, W. (2002). Dose-response modeling of continuous endpoints. *Toxicological Sciences*, 66:298–312.

Stokes, Z., Wong, W. K., and Xu, H. (2024). Metaheuristic solutions to order-of-addition design problems. *Journal of Computational and Graphical Statistics*, 33(3):1006–1016.

SWOG (2024). The twostage program. Accessed: 2024-08-28.

Tan, M. T. and Xiong, X. (2011). A flexible multi-stage design for phase II oncology trials. *Pharmaceutical statistics*, 10(4):369–373.

Torres-Saavedra, P. A. and Winter, K. A. (2022). An overview of phase 2 clinical trial designs. *International Journal of Radiation Oncology, Biology and Physics*, 112(1):22–29.

Wason, J., Magirr, D., Law, M., and Jaki, T. (2016). Some recommendations for multi-arm multi-stage trials. *Statistical methods in medical research*, 25(2):716–727.

Whitacre, J. M. (2011a). Recent trends indicate rapid growth of nature-inspired optimization in academia and industry. *Computing*, 93:121–133.

Whitacre, J. M. (2011b). Survival of the flexible: explaining the recent popularity of nature-inspired optimization within a rapidly evolving world. *Computing*, 93(2):135–146.

Wu, X., Wu, C., Liu, F., Zhou, H., and Chen, C. (2021). A generalized framework of optimal two-stage designs for exploratory basket trials. *Statistics in Pharmaceutical Research*, 13(3):286–294.

Xu, W., Wong, W. K., Tan, K. C., and Xu, J. X. (2019). Finding high-dimensional D-optimal designs for logistic models via differential evolution. *IEEE Access*, 7(1):7133–7146.

Yang, X.-S. (2010). *Engineering optimization: an introduction with metaheuristic applications*. John Wiley & Sons.

Zang, Y. and Lee, J. J. (2017). A robust two-stage design identifying the optimal biological dose for phase i/ii clinical trials. *Statistics in Medicine*, 36:27–42.

Zhan, T. (2024). *VisualizeSimon2Stage: Visualize Simon's Two-Stage Design*. R Foundation for Statistical Computing. Accessed: 2024-08-01.

Zhang, L. and Yuan, Y. (2020). Simon's two-stage design shiny app. Accessed: 2024-06-01.